13. Oscillations

IV Oscillations and Waves

Terms used

How displacement, velocity, acceleration varies with time

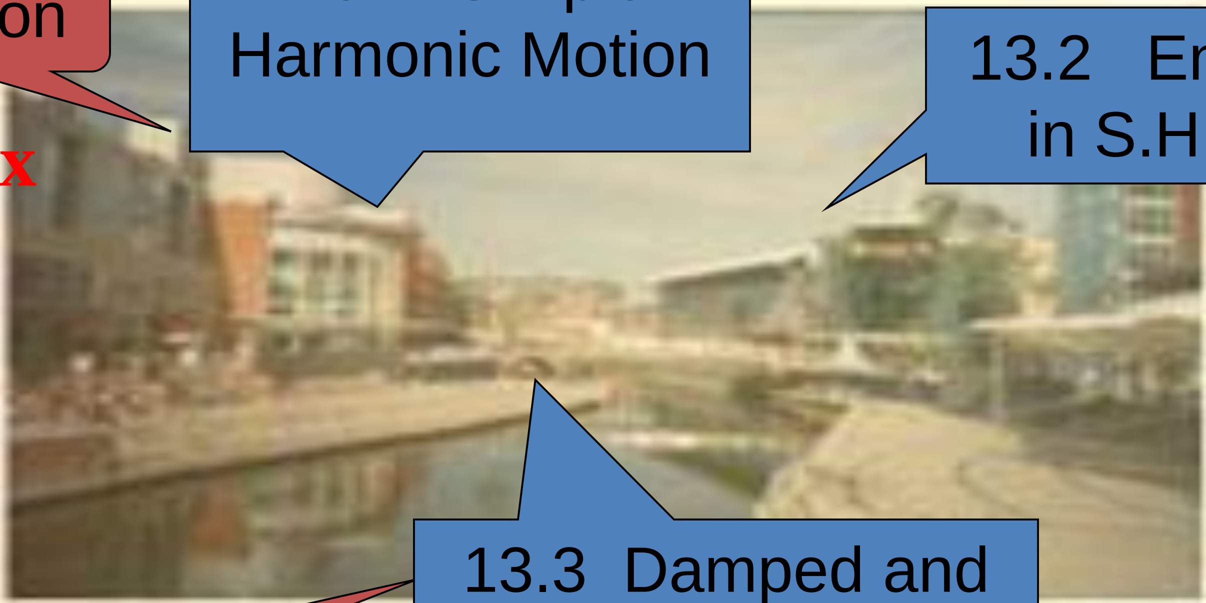
How velocity, acceleration varies with displacement

Definition

$\mathbf{a} = -\,\omega^2 . \mathbf{x}$

13.1 Simple Harmonic Motion

13.2 Energy in S.H.M.



13.3 Damped and Forced oscillations

Resonance

Simple Harmonic Motion

The Michael Whiteman Artrock Orchestra

http://surendranath.tripod.com

13.1 Simple harmonic motion

- Any motion that repeats itself after a certain period is known as a **periodic** motion, and since such a motion can be represented in terms of sines and cosines it is called a **harmonic** motion.

- **Simple harmonic motion** (s.h.m. for short) is the name given to a particular type of harmonic vibration. The following are examples of simple harmonic motion:

Examples

a test-tube bobbing up and down in water (Figure 1)
a simple pendulum
a compound pendulum
a vibrating spring
atoms vibrating in a crystal lattice
a vibrating cantilever
a trolley fixed between two springs
a marble on a concave surface
a torsional pendulum
liquid oscillating in a U-tube
a small magnet suspended over a horseshoe magnet
an inertia balance

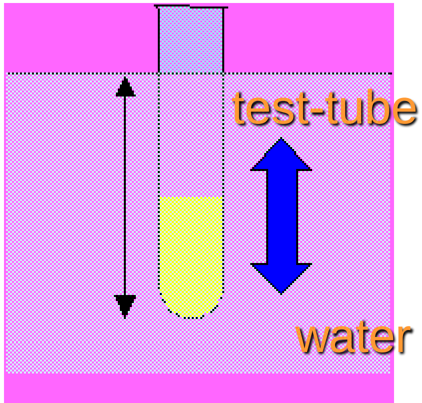
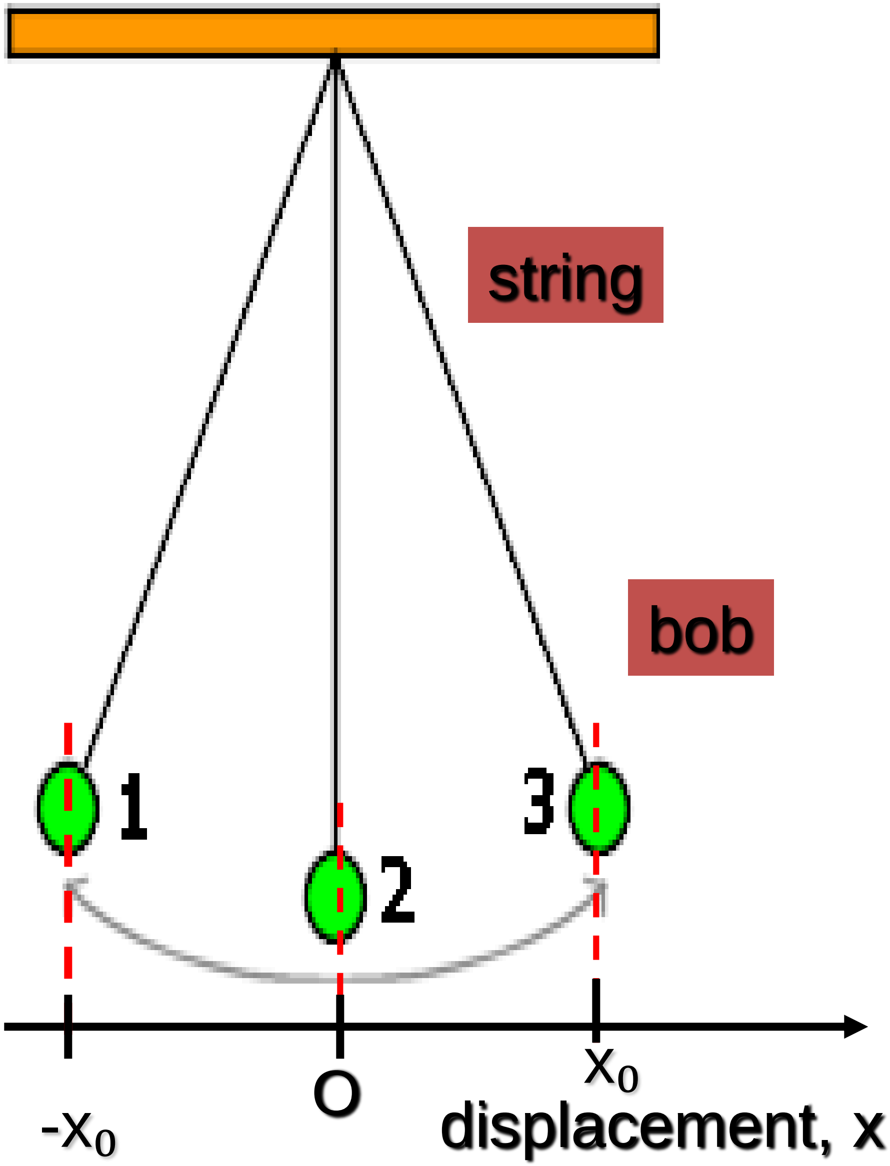


Figure 1

Example of free oscillation



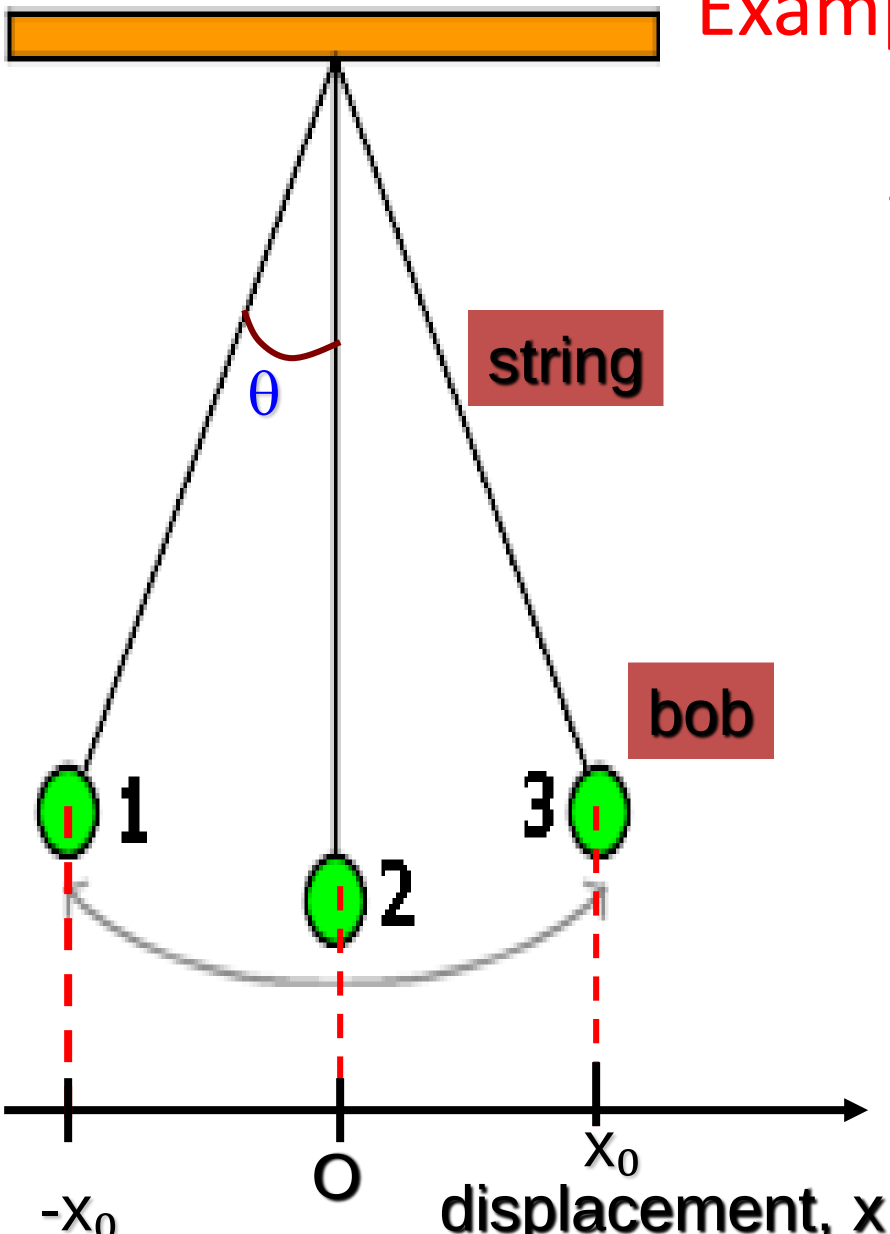
$$T = 2\pi\sqrt{\frac{L}{g}}$$

Natural frequency, $f_o = {}^1/_T$

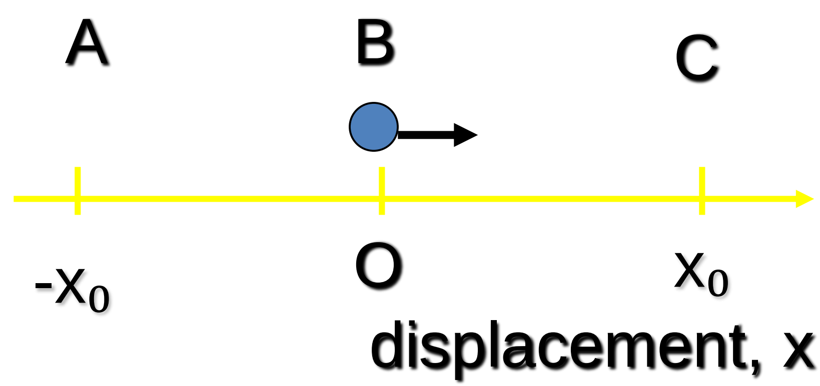
Oscillating simple pendulum. In free oscillation, when the bob is displaced it oscillates.

- The bob swings between two limits – the maximum and minimum displacement.
- The centre of the oscillating is the rest position, O.
- Rest position (equilibrium position) is when the pendulum is at rest.

Example of free oscillation

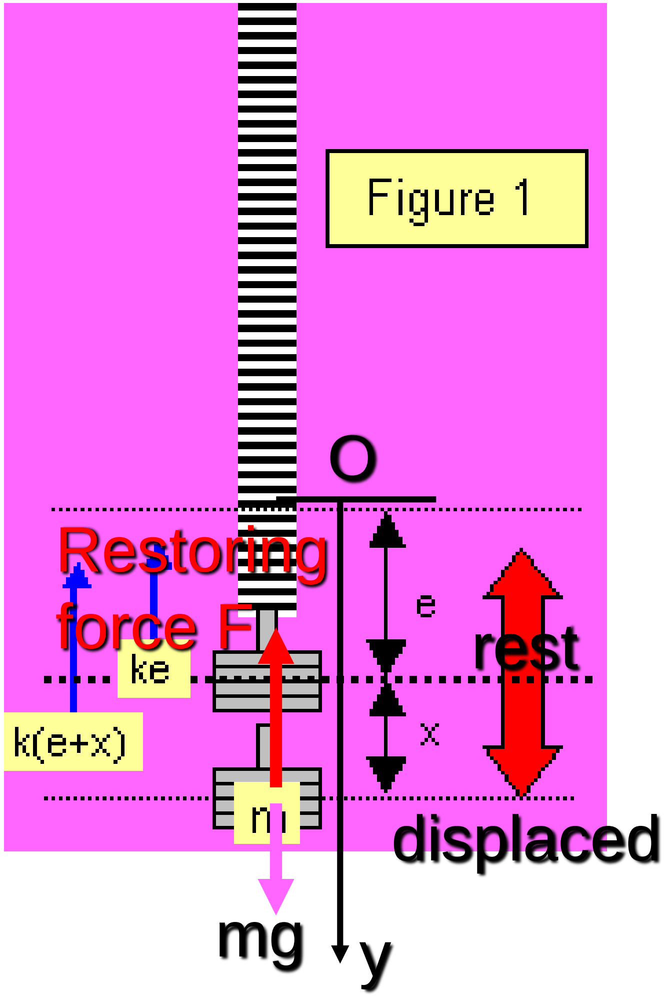


Suppose the oscillation starts from position 2 and moving to the right.



When $\theta < 10^{o}$, the motion is near to linear motion.

The helical spring



- Consider a mass m suspended at rest from a spiral spring and let the extension produced be *e*. If the spring constant is k, then mg = k*e* .....[1]

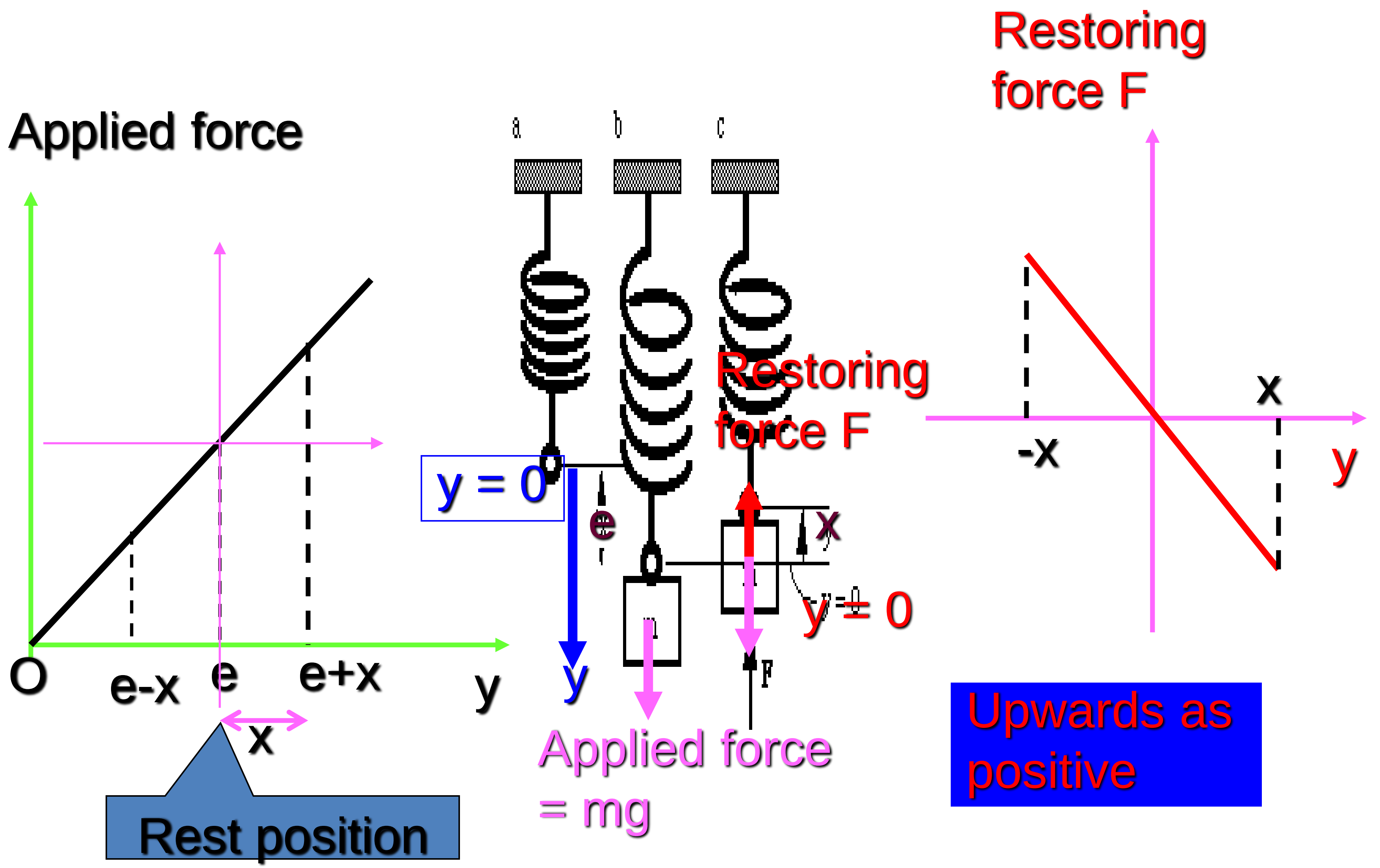
- The mass is then pulled down a small distance x and released. The mass will oscillate due to both the effect of the gravitational attraction (mg) and the varying force in the spring (k(*e* + x)).

At any point distance x from the midpoint:

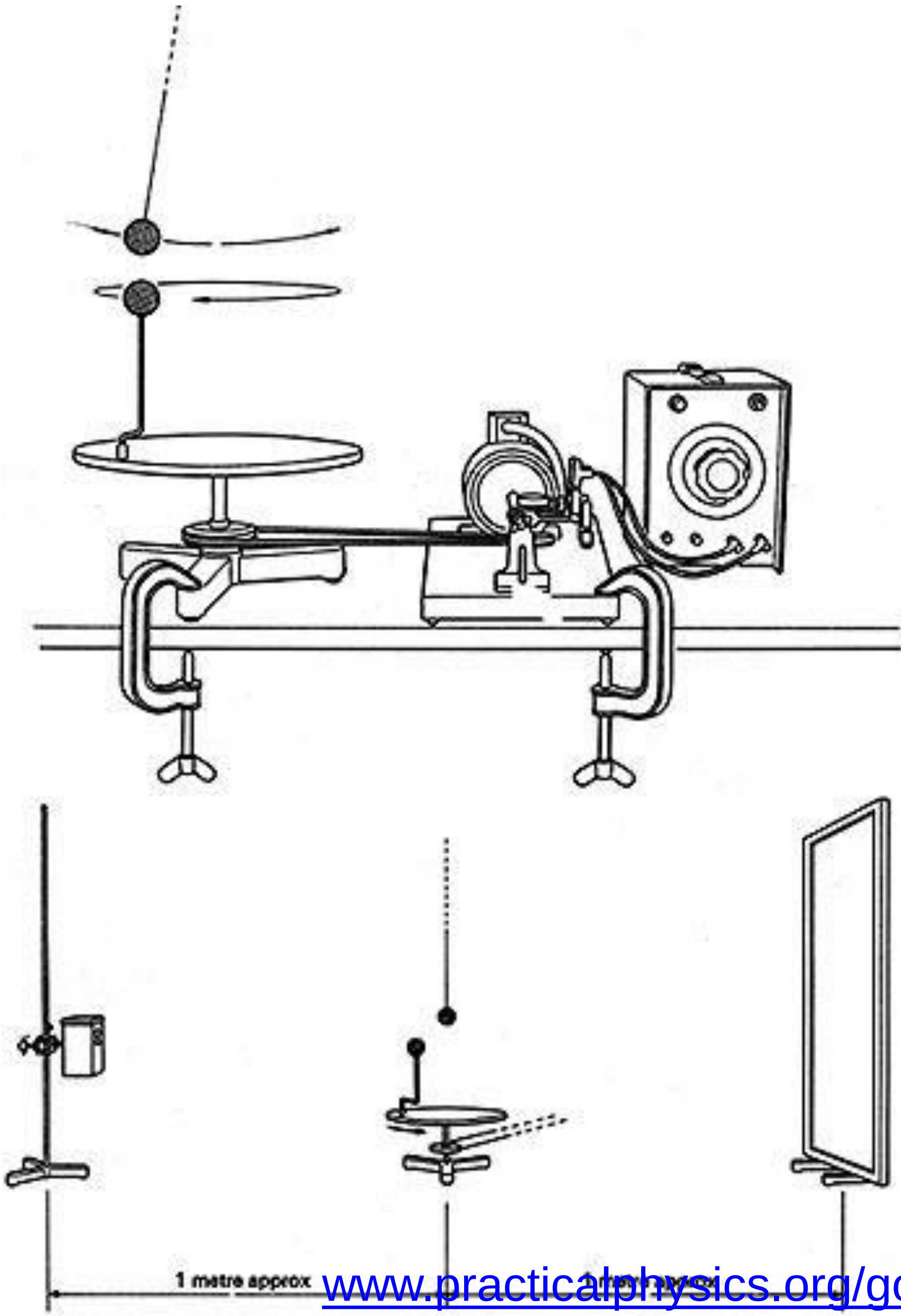
restoring force = k(*e* + x) – mg [2]

- ma = - kx (negative sign as F and x are in opposite direction)

Spring



Restoring force F
Applied force
a
b
c
Restoring force F
y = 0
e
x
y = 0
-x
x
y
O
e-x
e
e+x
y
y
x
Applied force = mg
Upwards as positive
Rest position

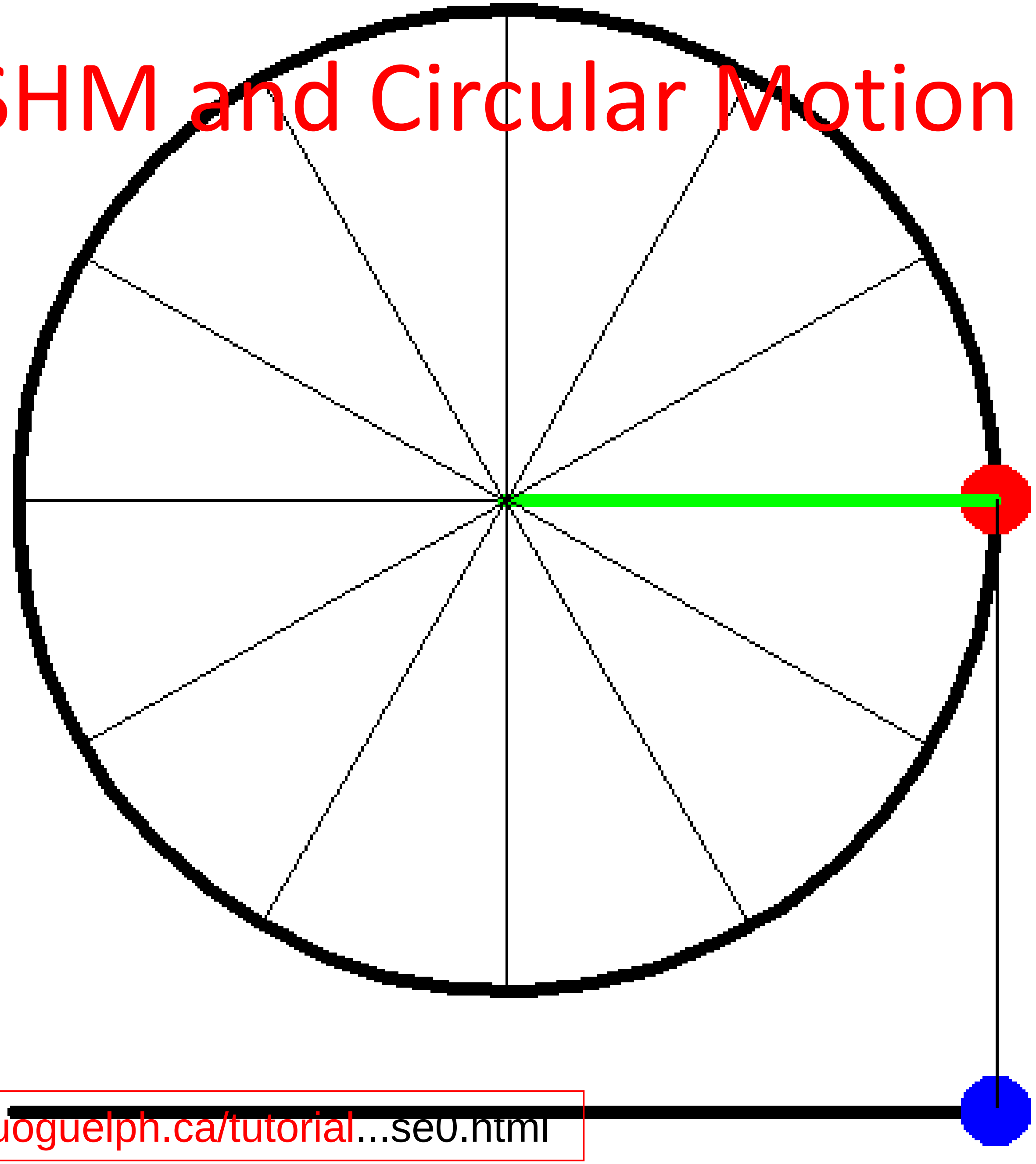


Circular motion and SHM

- As the ball moves with constant angular velocity in uniform circular motion the shadow of the ball on the screen performs SHM

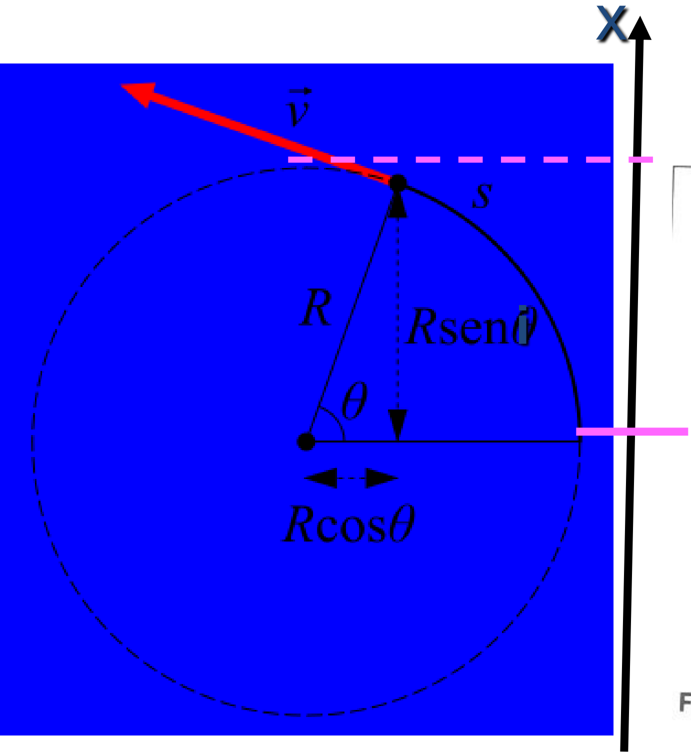
www.practicalphysics.org/go/Experiment_970.ht...

SHM and Circular Motion



www.physics.uoguelph.ca/tutorial...se0.html

Simple Harmonic Motion and uniform circular motion



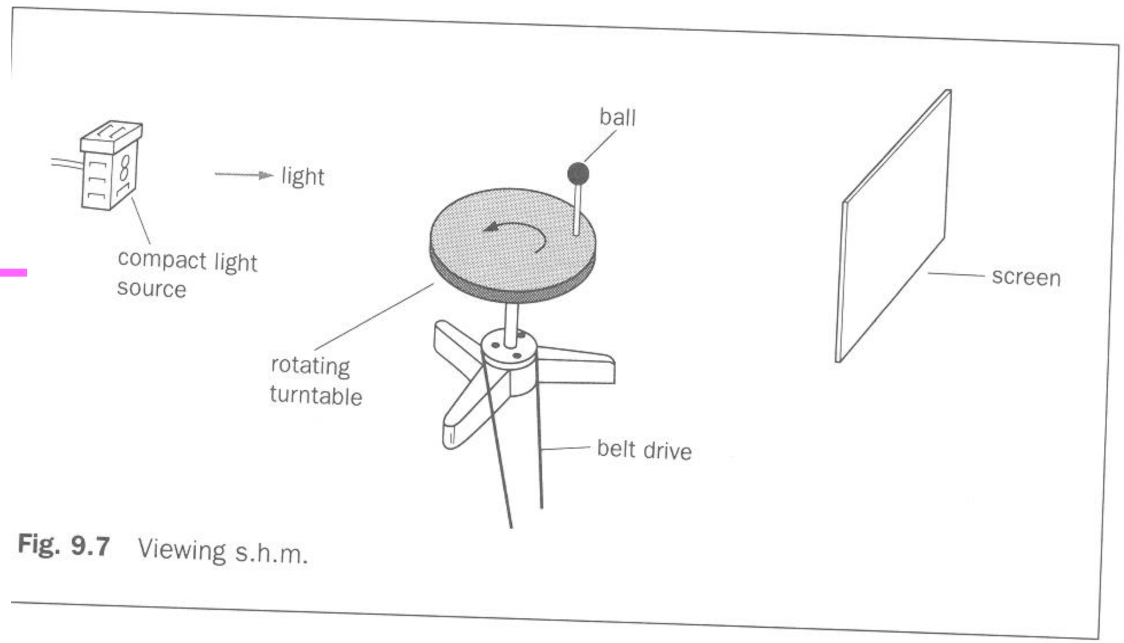


Fig. 9.7 Viewing s.h.m.

oscar.iitb.ac.in/AvailableAnimationByCategory...

Experiment.

- Set up the apparatus in a straight line in a darkroom.
- Switch on the light source, it cause the shadow of the ball to fall on the screen
- As the turntable rotates at constant angular velocity ω, the ball moves in a circle of radius x_o.
- The shadow of the ball moves in SHM in a straight with amplitude x_o.

Circular motion	**SHM**
Radius of circle, r	amplitude, x_o
angular velocity, ω	angular frequency, ω
uniform speed, $v_o = rw$	maximum speed, $v_o = x_o\omega$
acceleration. $a_o = r\omega^2$	max. acc. $a_o = -\omega^2 x_o$
one revolution	one oscillation

EXPERIMENT

Student investigation

The oscillation of a trolley fixed between two rigid supports by two springs is s.h.m. Using the apparatus shown in Figure 3, find out whether this is true. Using a sensor the variation of displacement and velocity with time may be found. Plot graphs of the values that you obtain. Do they agree with the results that you would expect?

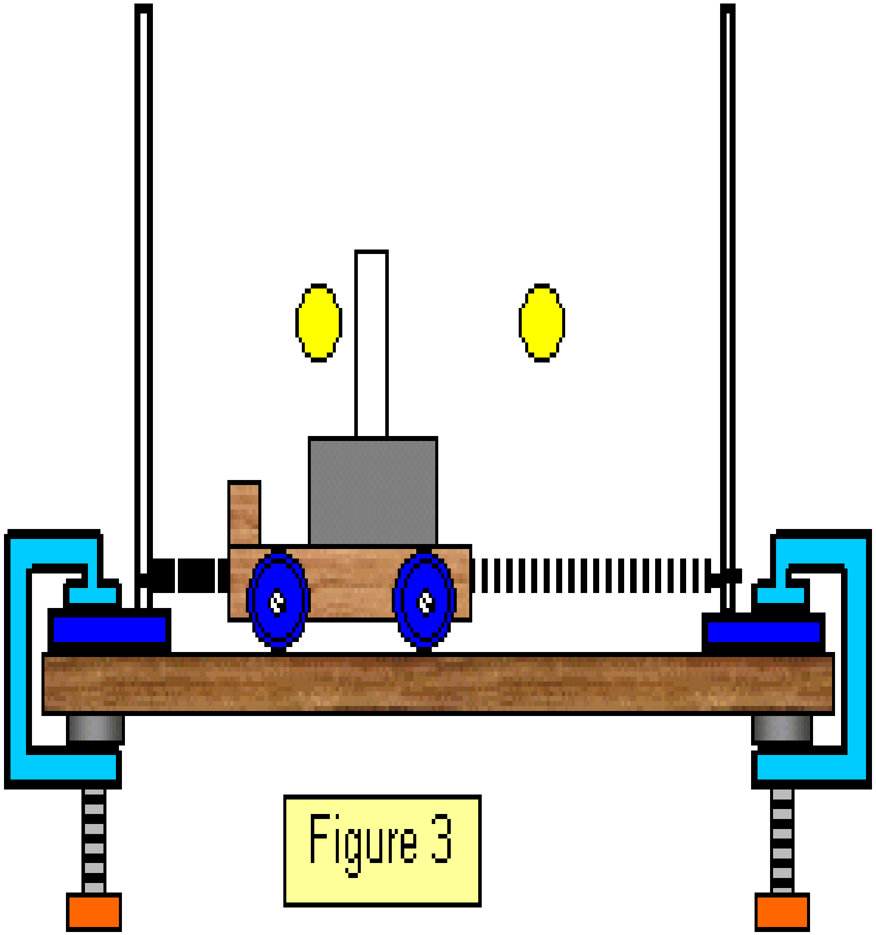


Figure 3

Experiment

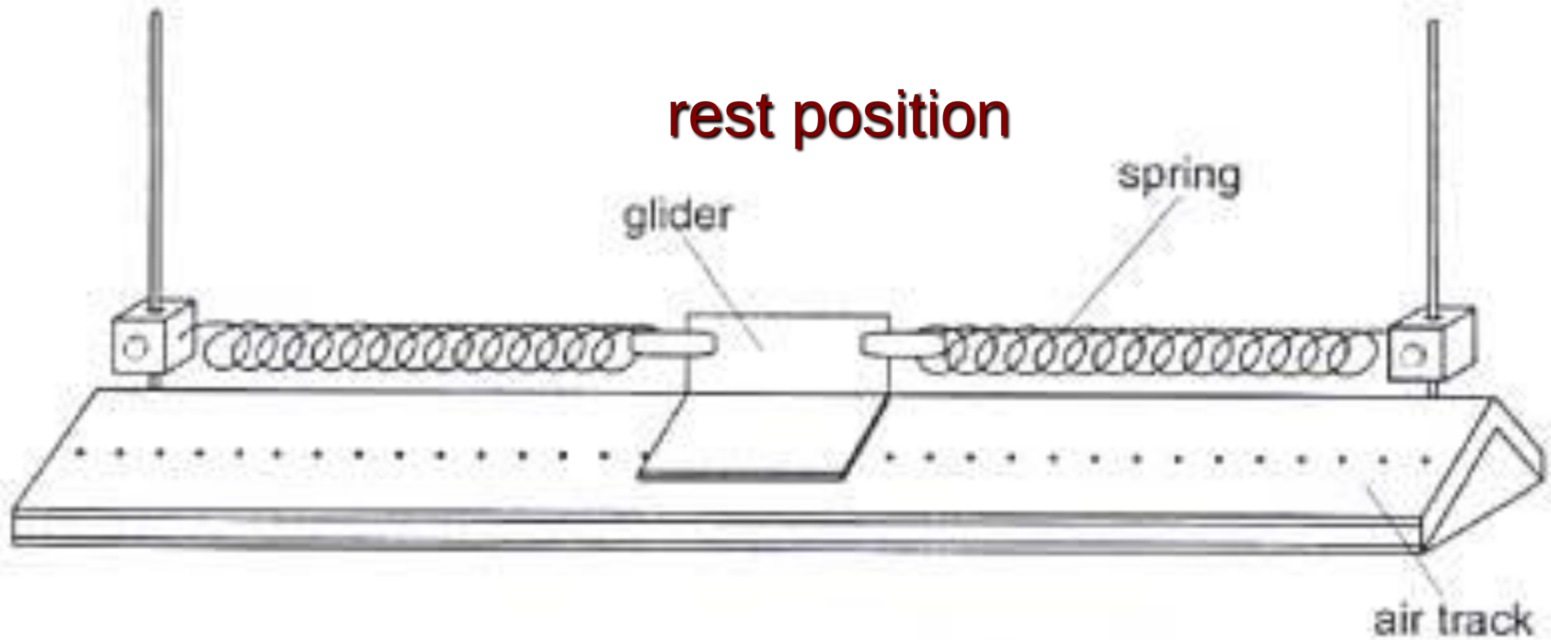


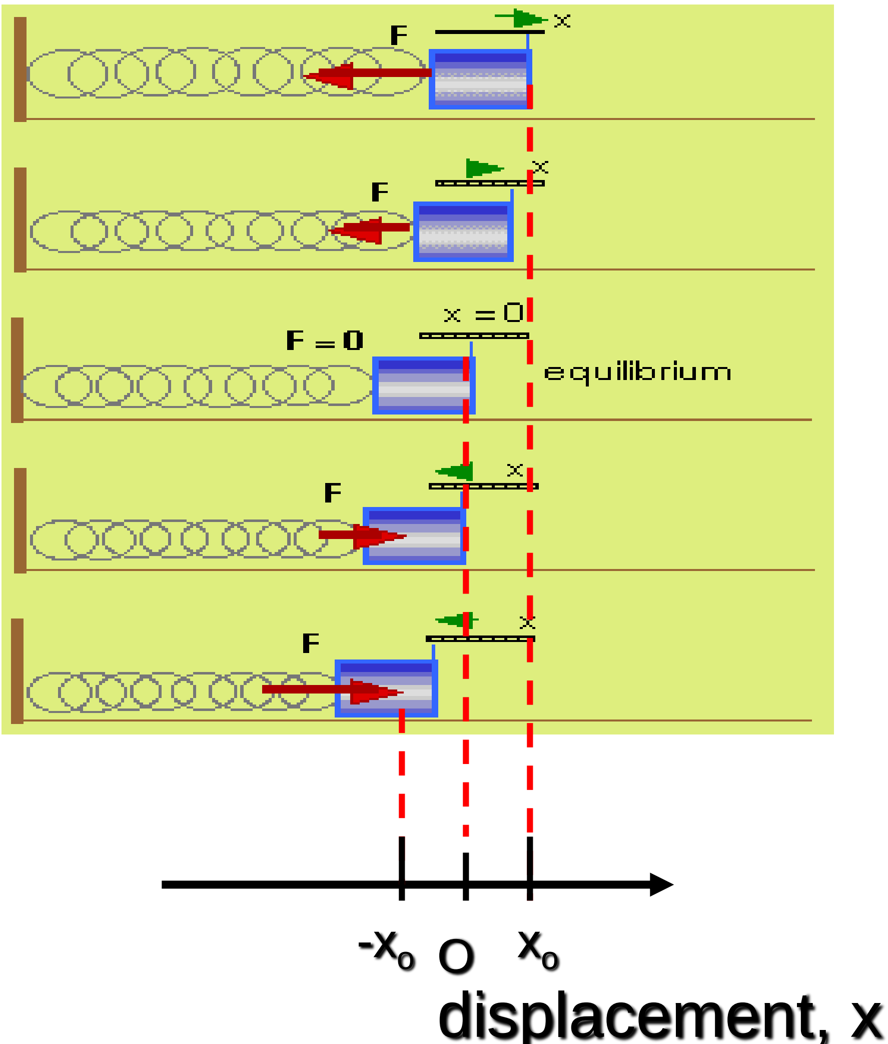
Fig. 2.1

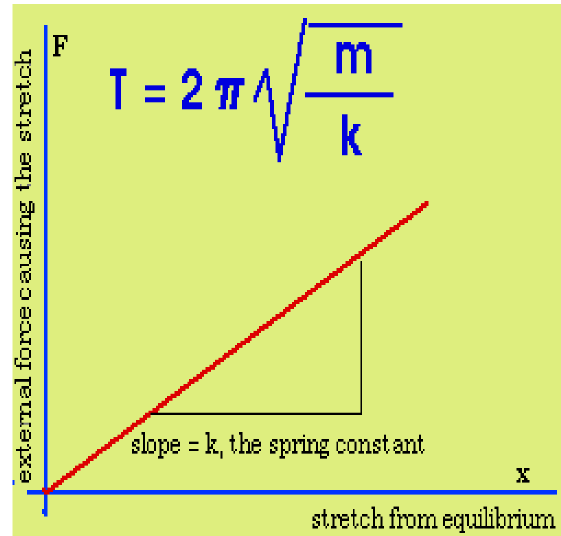
- When the glider is displaced it undergoes SHM

Example of free oscillation:

F is the restoring force

Natural frequency, $f_o = 1/T$





- Object attach to a spring oscillating horizontally on a smooth surface.
- F is the load on the spring i.e. mg

Terminology (Refer Slide 6)

One oscillation or one cycle is one complete to and fro motion about the centre point e.g. from A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ B $\rightarrow$ A or B $\rightarrow$ C $\rightarrow$ B $\rightarrow$ A $\rightarrow$ B.

Period (T) of an oscillation is the time to complete one oscillation.

Frequency (f, ν (Gk nu)) of oscillation is the number of complete oscillations per unit time.

- Unit: hertz (Hz), or cycles per s or s^{-1}.
- $f = {}^{1}/_{T}$

Amplitude (x_o, A, r) is the maximum displacement of the oscillator from the rest position.

Rest position (equilibrium position) is the position of the object when it is not oscillating.

Terminology

- **The *amplitude* is the maximum distance the mass moves from its equilibrium position. It moves as far on one side as it does on the other.**
- **The *time* that it takes to make one complete repetition or cycle is called the *period* of the motion. We will usually measure the period in seconds.**
- ***Frequency* is the number of cycles per second that an oscillator goes through. Frequency is measured in "hertz" which means cycles per second.**
- ***Period* and *frequency* are closely connected; they contain the same information.**

$$T = 1/f$$
$$f = 1/T$$

Terminology

Angular frequency (ω, Gk omega) of an oscillatory motion is frequency expressed in radians per second.

- $\omega = 2\pi f$ where f is the frequency of oscillation

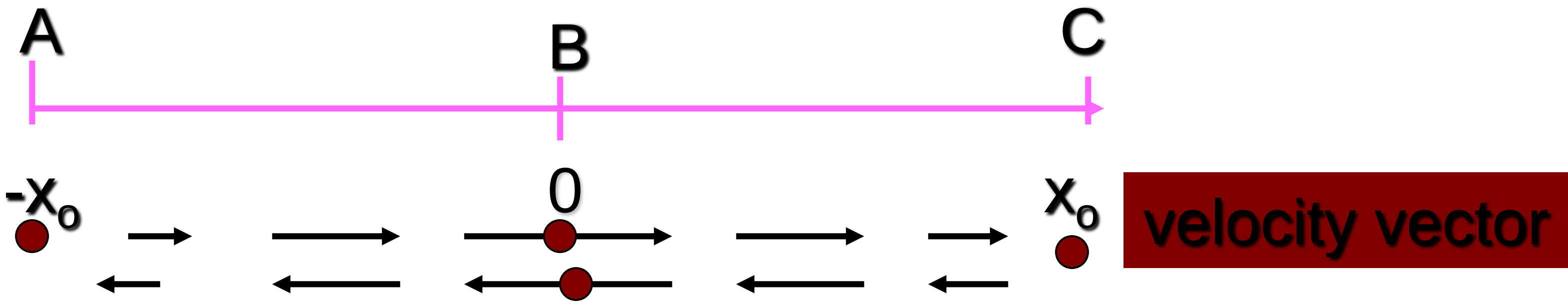
Displacement in s.h.m $x = r \sin \omega t$

Velocity and acceleration: $v = r\omega \cos \omega t$ and $a = - r\omega^2 \sin \omega t$

r = amplitude

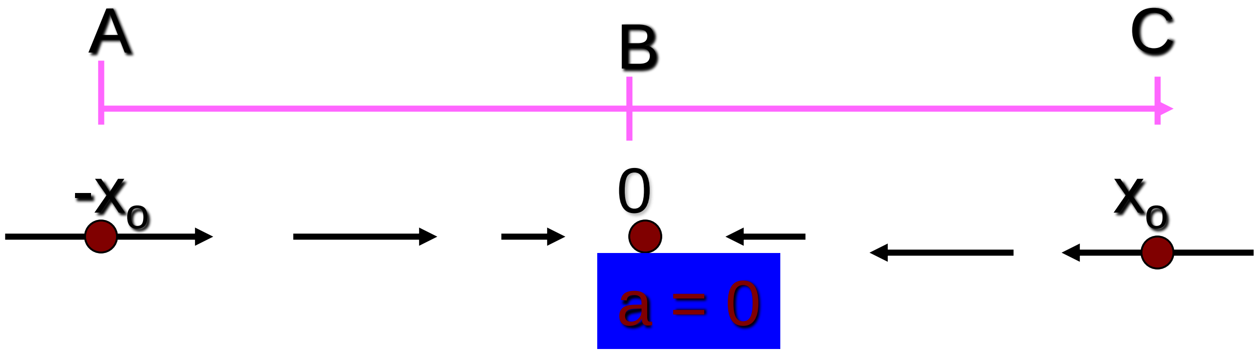
v is differential of x
A is differential of v

Velocity



- At the rest position (B) the velocity is maximum and moving to the right.
- The load undergoes deceleration (negative acceleration) as it moves towards its maximum amplitude © and its velocity must be zero at maximum amplitude where the retardation is maximum.
- The load then reverses direction and accelerates to rest position where the acceleration must be zero and the velocity is maximum but moving towards the negative x-direction.

Acceleration



- At zero displacement (position B) the acceleration is zero and at maximum displacement (position C) the acceleration is maximum but in opposite direction to displacement.
- At negative displacement (position A) the acceleration on the load is towards the rest position i.e. oppose to the displacement.

Variation of displacement, velocity and acceleration with time

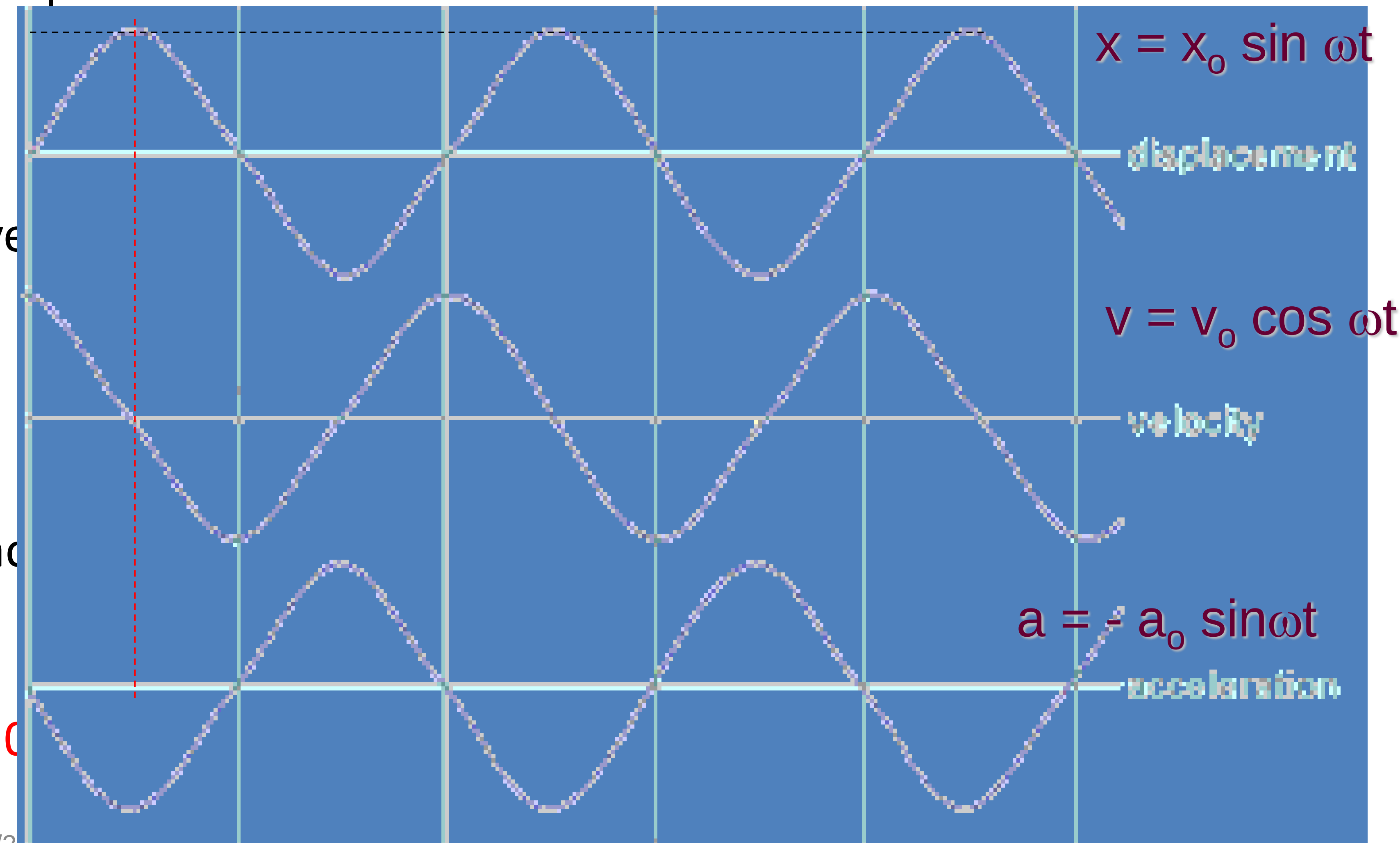
Displacement

r

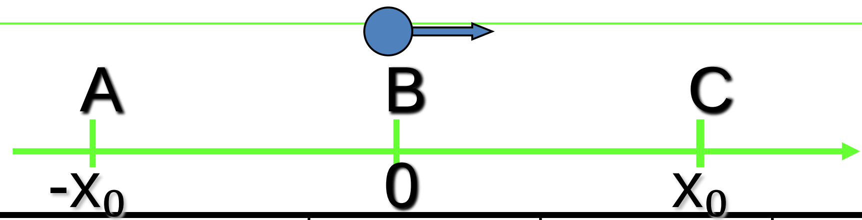
ve

ac

0



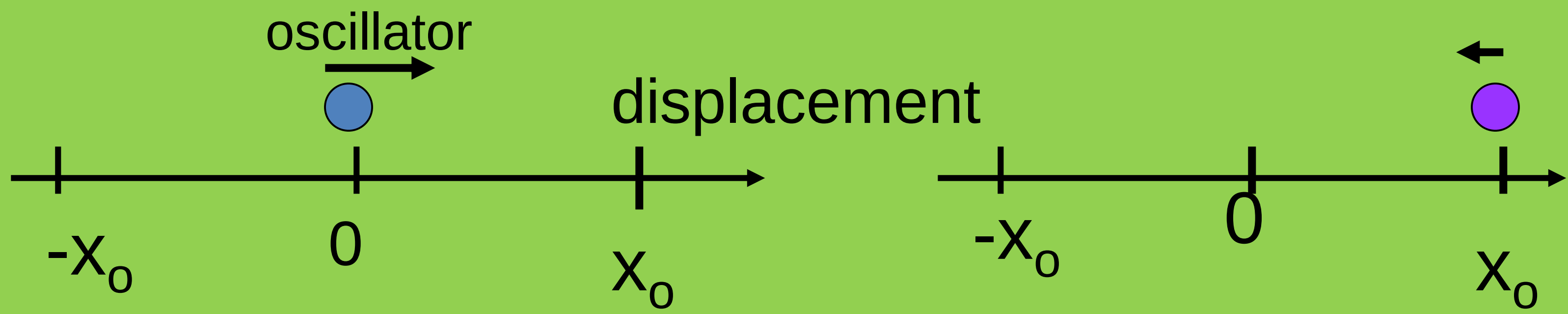
Variation with displacement and time



to the right is positive

Position	B	C	B	A	B
Time	0	¼ T	½T	¾T	T
displacement	0	x_o	0	$- x_o$	0
Velocity	v_o	0	$-v_o$	0	v_o
Acceleration	0	$- a_o$	0	a_o	0
Kinetic energy	E_{ko}	0	E_{ko}	0	E_{ko}
Potential energy	0	E_{po}	0	E_{po}	0

Alternative equation



oscillator= oscillating body

For an oscillating object starting from 0(the rest position) and moving in the positive x-direction, the displacement at any time is given by

Equation of oscillator,

$x = x_o \sin \omega t$

Equation of oscillator,

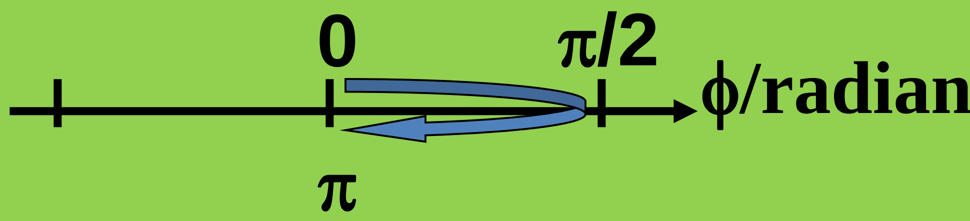
$x = x_o \cos \omega t$

If the oscillator starts oscillating at C (the maximum displacement in x-direction) and moving towards the rest position then the phase is $\pi/2$ radian.

$x = x_o \sin (\omega t + {}^{\pi}/_{2})$

$= x_o \cos \omega t$

General Equation



General equation of an oscillator in SHM:

$x = x_o \sin(\omega t + \phi)$

where ϕ is the phase in radian

- **the phase locate the position of particle initially.**

(to date there is no question in A-level with phase difference. However, you are required to know that sin can be written as –sin, cos and –cos. With this substitution's, you already include phase difference).

Mathematics corner [info]

$x = x_o \sin \omega t$ .... [1]

$v = v_o \cos \omega t$ .....[2]

Squaring and adding the equations

$$\frac{x^2}{x_o^2} + \frac{v^2}{\omega^2 x_o^2} = 1$$

as $v_o = x_o \omega$

and $\sin^2 \omega t + \cos^2 \omega t = 1$

This an equation of an ellipse.

$$v = \pm \omega \sqrt{x_o{}^2 - x^2}$$

Total mechanical energy,

$E = E_k + E_p$

Kinetic energy

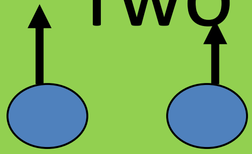
$E_k = ½ mv^2$

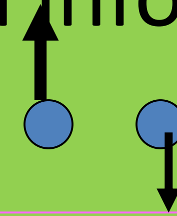
$= ½ m\omega^2(x_o{}^2 - x^2)$

$= E - ½ m\omega^2 x^2$

$E_p = ½ m\omega^2 x^2$

Two oscillating particle (additional info)





In phase. When two oscillators are oscillating in phase, both object oscillate in the same direction, reaching maximum displacement or minimum displacement at the same time i.e. oscillating with the same frequency.

Two oscillator are oscillating in ***antiphase*** or π radian out of phase, if one is moving upwards from the rest position while the other is moving downwards from rest position i.e. one will be at positive amplitude while the other will be at negative amplitude at the same time.

Definition

Simple harmonic motion (SHM) is defined as a motion in which the acceleration of a body

a) is proportional to its displacement from a fixed point and

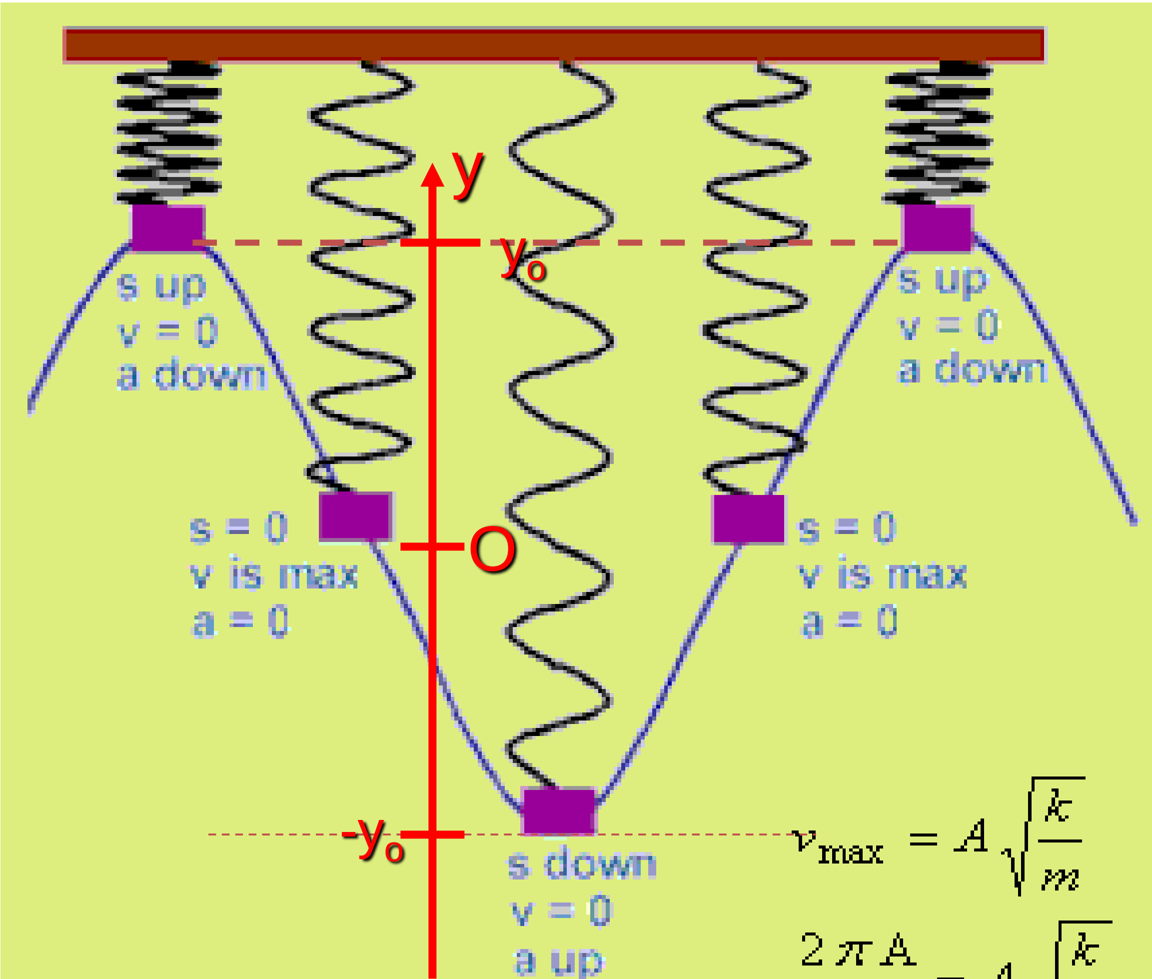
b) is always directed towards that point.

$$\mathbf{a} = -\,\omega^2.\mathbf{x}$$

Solution to the equation: $x = x_o \sin \omega t$ or $x = x_o \cos \omega t$

- The body is oscillating equal distances either side of some fixed point. Amplitude of oscillation is constant.
- No resistive forces to oppose the motion (undamped).
- Total energy is constant.

Example of free oscillation

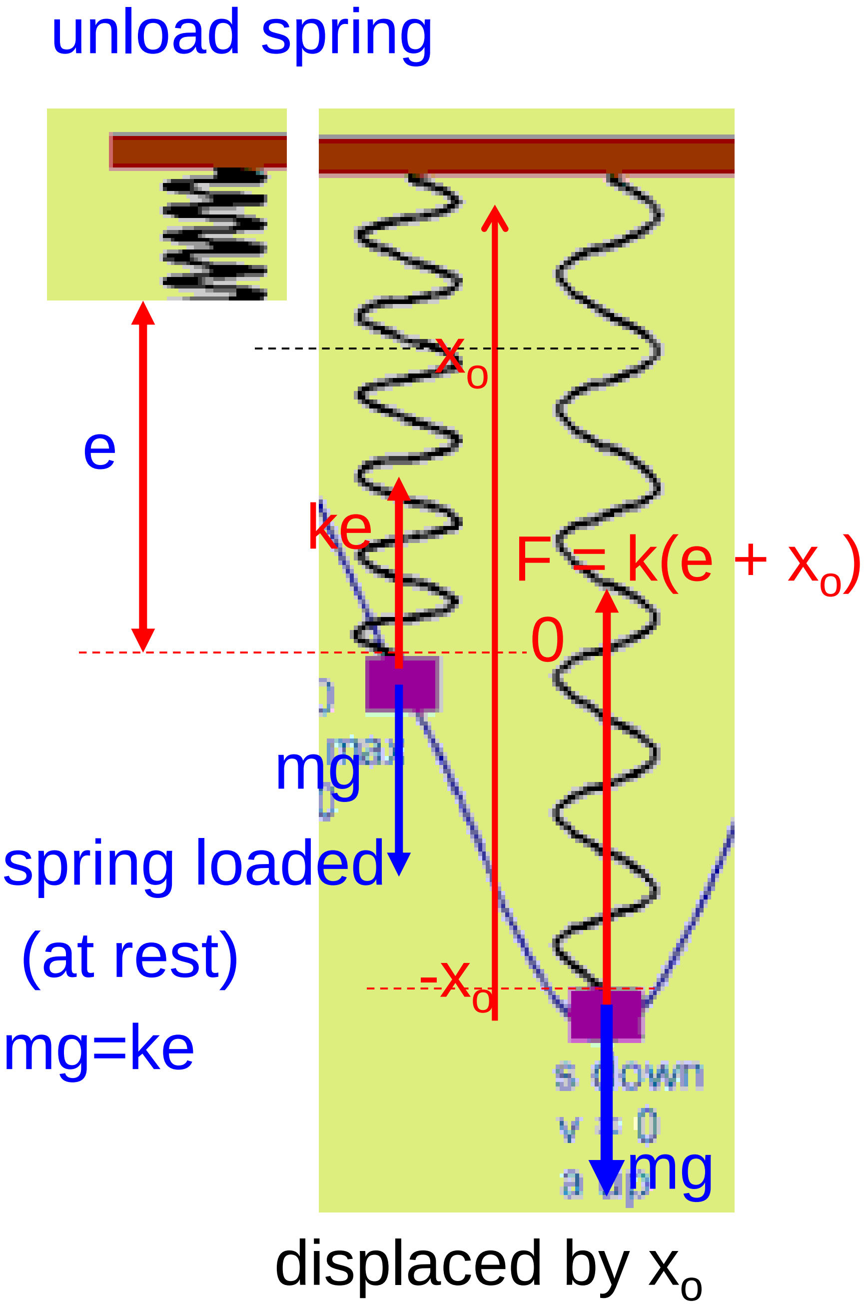


$$v_{max} = A\sqrt{\frac{k}{m}}$$

$$\frac{2\pi A}{T} = A\sqrt{\frac{k}{m}}$$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

- A load attach to a spring at one end and the other end fixed.
- When displaced the load oscillates between two limits about the rest position.



Vibrating spring

When the spring is displaced a distance x_o from rest position, the resultant force upwards (x_o <e)

$F_R = F - mg$

$ma = -[(ke + kx) - mg]$

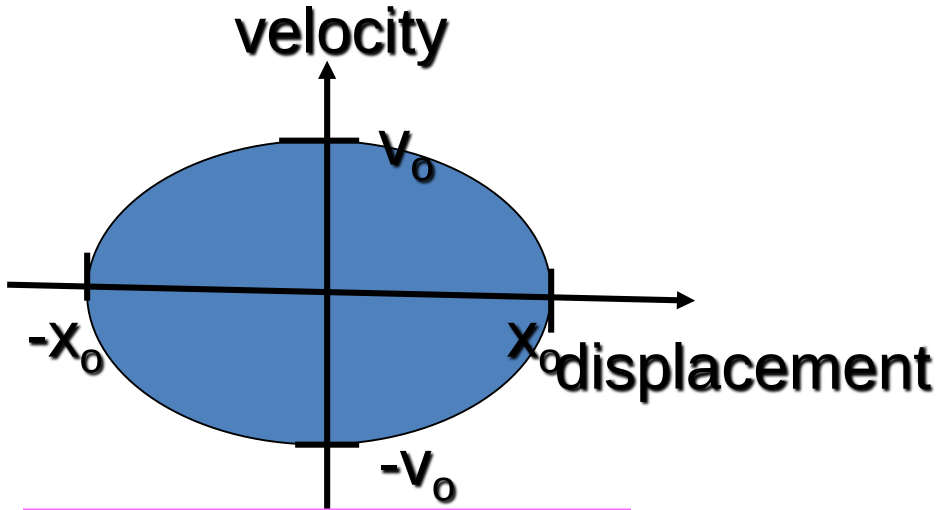
negative sign as the acceleration is opposite to the displacement.

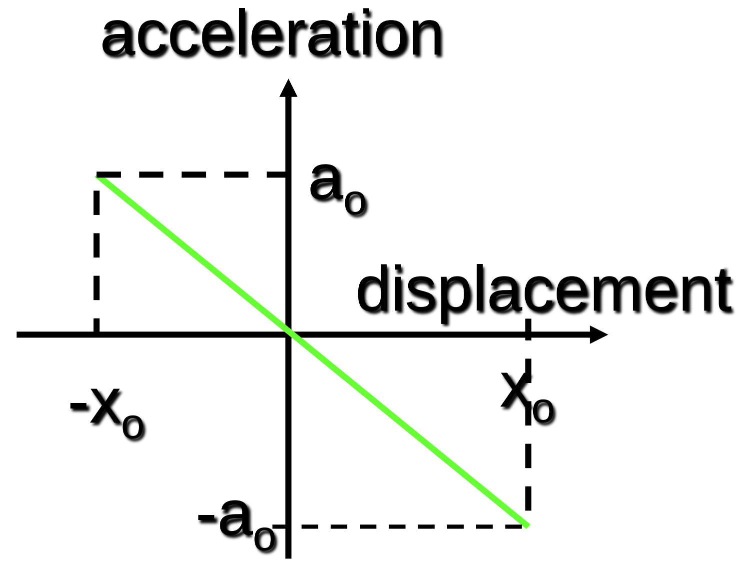
Replacing ke by mg, then

$$a = -\frac{k}{m}x$$

we can see that the *a* is proportional to *x*

Variation of velocity and acceleration with displacement.





$$v = \pm\,\omega\sqrt{x_o{}^2 - x^2}$$

Acceleration

$a = -\ \omega^2\ x$

(y = m x)

Equation of an ellipse (info)

$$\frac{y^2}{b^2} + \frac{x^2}{a^2} = 1$$

$$\frac{v^2}{(\omega x_o)^2} + \frac{x^2}{x_o{}^2} = 1$$

Try yourself (info.)

Show that if the acceleration of an oscillator is given as, $a = -\omega^2 x$, then the velocity of the oscillator is

$$v = \pm\omega\sqrt{x_o^{\ 2} - x^2}$$

Derivation:

$$a = \frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}$$

boundary condition, x=0, $v=v_o$ and at any displacement, x the velocity is v.

$$\int_{v_o}^{v} v dv = -\omega^2 \int_0^x x.dx$$

$v^2 - v_o^{\ 2} = -\omega^2 x^2$ and $v_o = \omega x_o$

Example 14.0

A pendulum takes 50.0 s to complete 20 oscillations. Calculate

a) the period,

b) the frequency

c) the angular frequency,

(Ans. a) 2.5 s b) 0.40 Hz, c) 2.5 rad. s^{-1})

Solution

a) T= 50/20 = 2.5 s

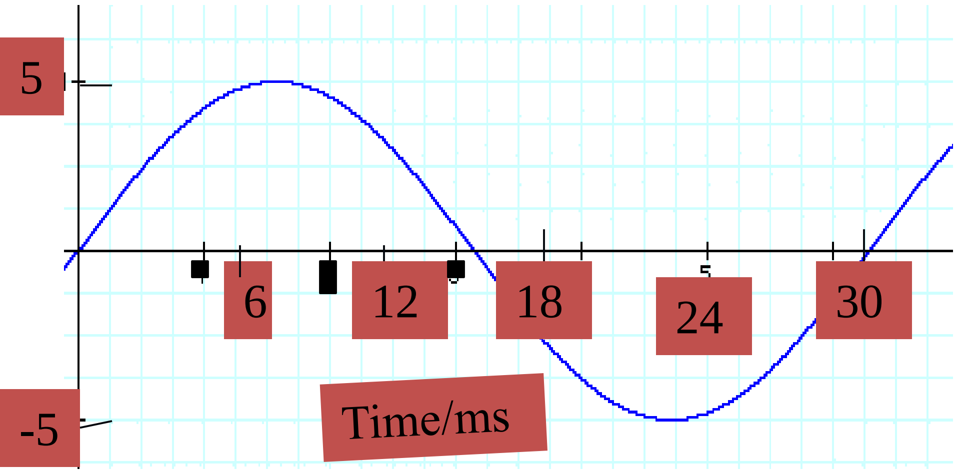
b) f = 1/T =0.4 Hz

c) ω=2πf = 2π(0.4)

= 2.5 rad/s

Example 14.1

displacement/cm



The displacement of an oscillating object is given by the graph below.

Find a) the amplitude

b) the period,

c) the frequency of the oscillation.

d) Angular frequency

State the equation of the oscillating object.

a) 5 cm

b) period = 30 ms

c) $f=1/30\times10^{-3}$ = 33.3 Hz

d) $\omega= 2\pi f$ = 209 rad/s

e) x=(5/cm)sin 209t

Example 14.2

The displacement of an oscillator is given in cm by

$x = 20 \sin 4\pi t$.

Find

a) the amplitude,

b) the angular frequency,

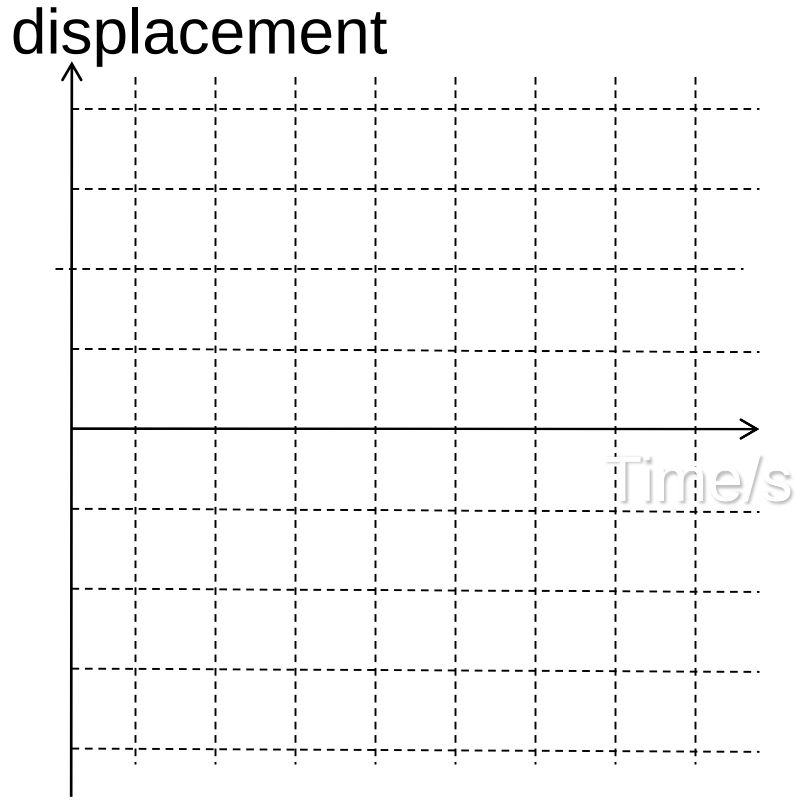
c) the period of oscillation,

d) the maximum magnitude of velocity,

e) the displacement at the time of 0.20 s.

Sketch the displacement-time graph of the motion.

(Ans: a) 20 cm; b) 12.6 rad/s; c) 0.50 s; d) 2.52 m/s; e) 11.8 cm)



Solution

a) 20 cm
b) $\omega t = 4\pi t$, so $\omega = 4\pi$ rad/s
c) $T = 2\pi/\omega = 0.5$ s
d) $v = r\,\omega = 20(4\pi)$
   $= 80\pi$ cm/s
e) $x = 20 \sin 4\pi[0.2]$
   $= 11.8$ cm

Maths corner:

$\cos 0 = 1$

$\cos 180 = -1$

$\sin(A+B)$

$= \sin A \cos B + \cos A \sin B$

$\sin(\theta+180) = -\sin\theta$

$\sin(\theta+90°) = \cos\theta$

Example 14.3

$$T = 2\pi\sqrt{\frac{m}{k}}$$

A load of 70.0 N causes the spring to extend by 5.0 cm. calculate,

a) the spring constant,

b) the load required for the spring to extend to 7.0 cm.

Solution

a) $k = 70/0.05 = 1400\ \text{N m}^{-1}$

b) $F = kx = 1400(0.07)$

$= 98\ \text{N}$

c) $= 0.449\ \text{s}$

$$T = 2\pi\sqrt{\frac{70/9.81}{1400}}$$

When the spring is oscillating with a load of 70 N calculate,

c) the period,

d) the angular frequency,

e) the natural frequency of oscillation of the spring.

(Ans. a) $1400\ \text{N m}^{-1}$, b) 98 N, c) 0.449 s, d) $14\ \text{rad.s}^{-1}$ e) $2.23\ \text{s}^{-1}$)

Solution

d) $\omega = 2\pi/0.449 = 14\ \text{rad/s}$

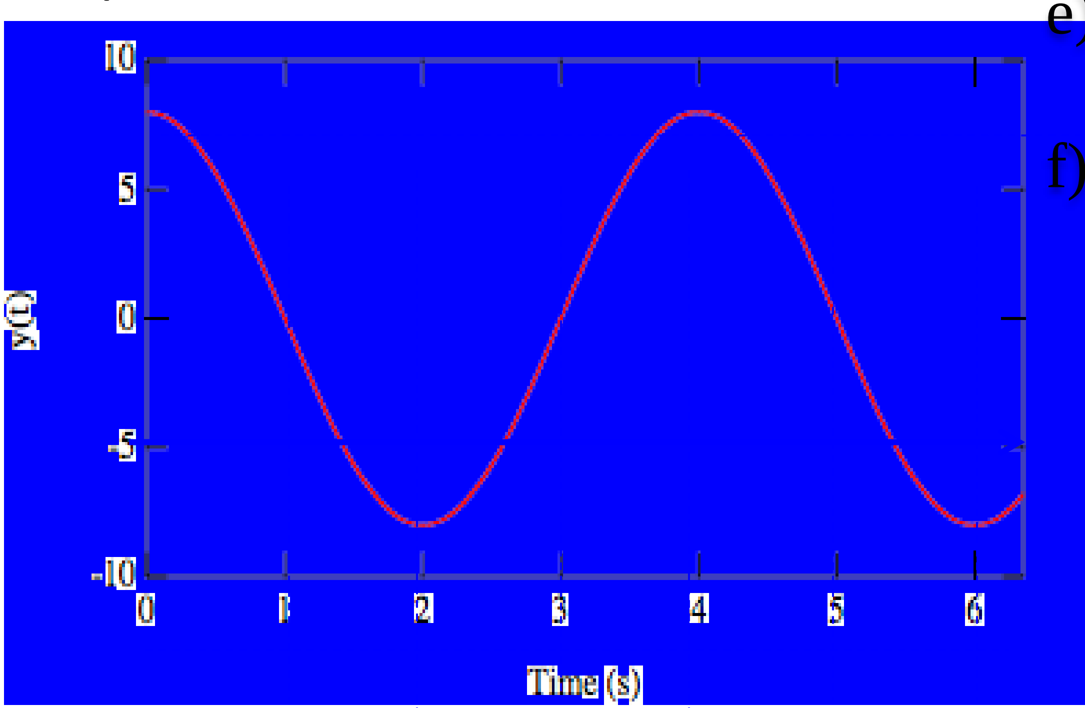
e) $f = 1/T = 1/0.449$

$= 2.23\ \text{s}^{-1}$

Example 14.4

The displacement of an oscillating object is given by the graph below.

Displacement/cm



Find a) the amplitude

b) the period,

c) the frequency of the oscillation.

d) Angular frequency

e) State the equation of the oscillating object.

f) Sketch the velocity-time graph of the oscillation.

a) 8 cm

b) 4 s

c) f = ¼ = 0.25 s^{-1}

d) $\omega = 2\pi/4 = 1.57$ s

e) y = (8/cm)cos 1.57t

(a) State the equation defining simple harmonic motion. [1]

$a = -\omega^2 . x$

(b) The graph, Fig. 16, shows how the accelera… object undergoing simple harmonic motion varies with time.

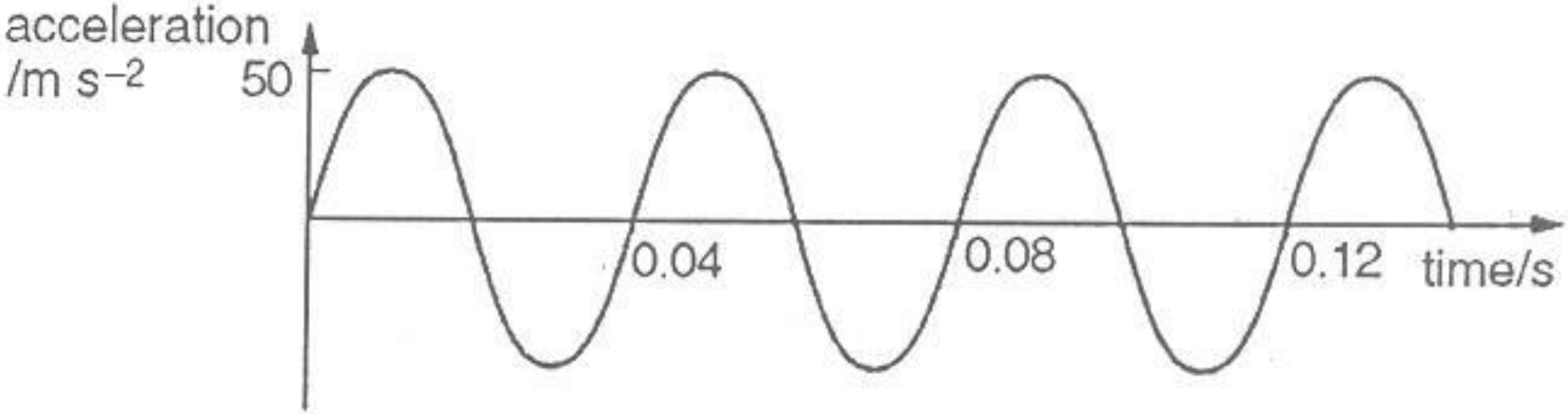


Fig. 16

Deduce, from the numerical values given on the graph, the values for this simple harmonic motion of

(i) the period, 0.04 s

(ii) the frequency,

(iii) the angular frequency ω,

(iv) the amplitude x_0 of the oscillation. [6]

(c) Sketch on Fig. 17 a graph which shows how the displacement varies with time.

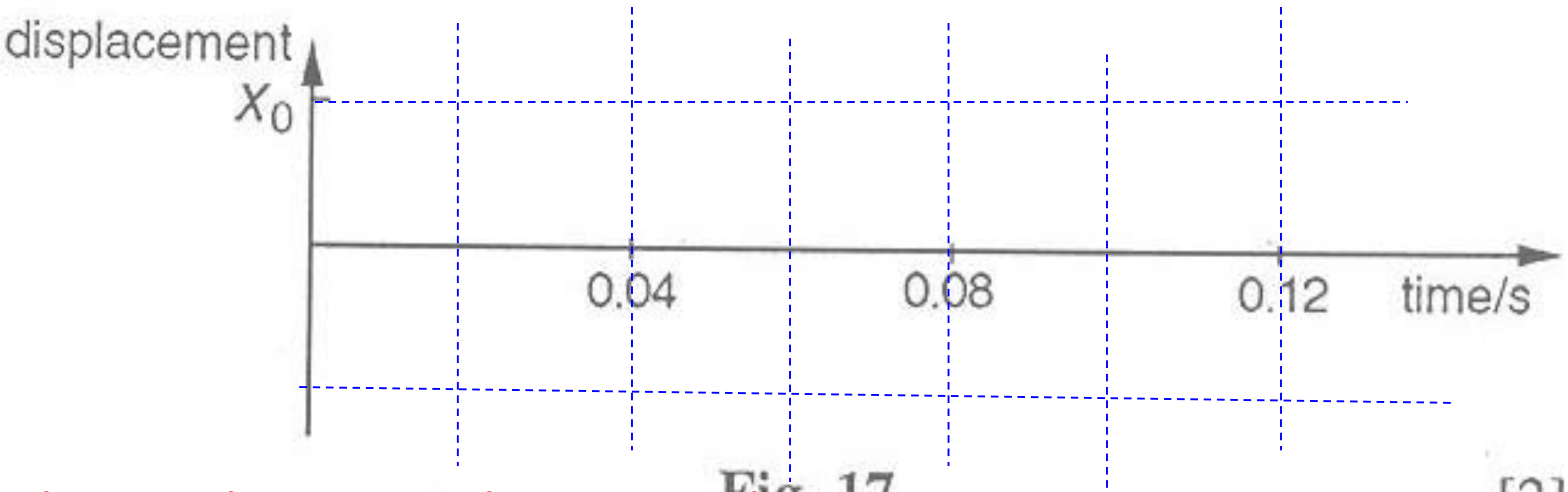


Fig. 17

[2]

N93/II/3

(Ans. bi) 0.04 s; ii) 25 Hz; iii) 157 rad/s; iv) 2.03×10^{-3}m

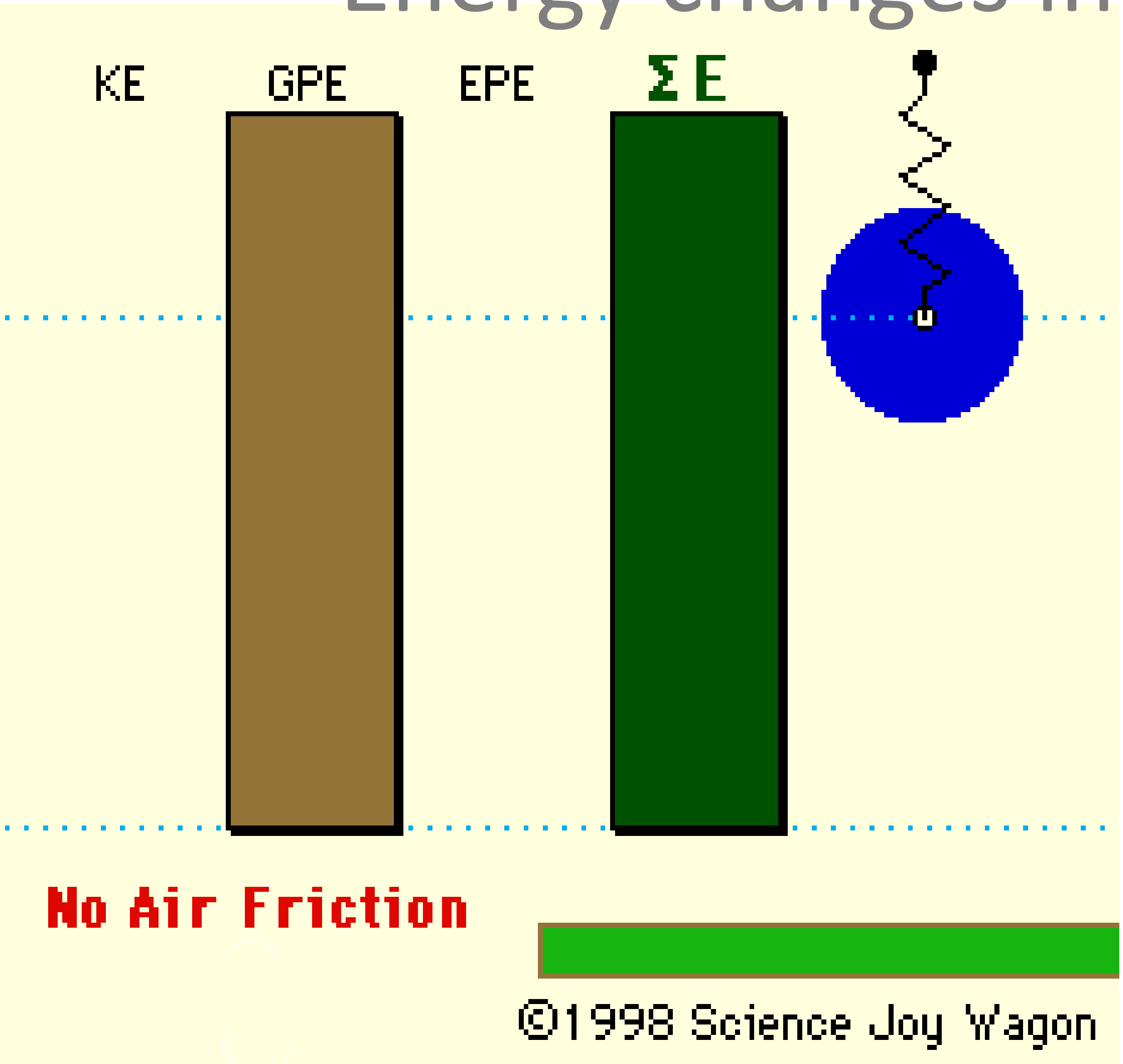
Review

- Able to relate circular motion with SHM
- Recall s = r θ
- Define I radian
- Recall v = 2πr/T : v = rω and ω = 2π/T
- Can you come up with appropriate SHM equation for given questions.
- Recall $a = -\omega^2 x$ and $v = \pm\omega\sqrt{x_o^2 - x^2}$
- How to calculate maximum acceleration and maximum velocity.
- Define SHM
- Graphs : displacement – time, velocity – time and acceleration – time
- Graphs : Force – displacement, acceleration displacement (both with ω constant)
- Graphs : velocity – displacement with ω changing.

PYQ

- 1. M/J 02 – 4a
- 2. O/N 03 – 2
- 3. M/J 05 – 4
- 4. O/N 05 – 4
- 5. M/J 06 – 4

Energy changes in SHM



http://physicsquest.homestead.com/questSHM.html

- GPE = gravitational PE
- EPE = elastic PE

Energy in simple harmonic motion

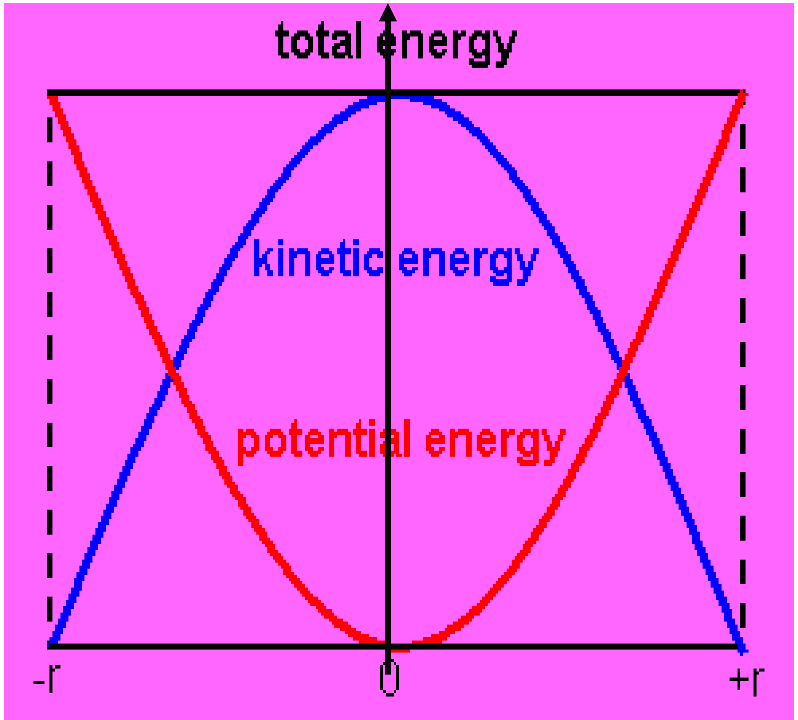


Figure 1

For a spring,

The energy stored (PE)

$= \frac{1}{2} k x^2$

where k = spring constant

Total energy,

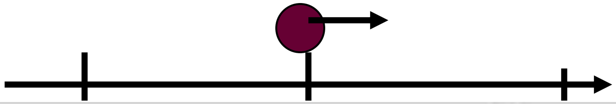
$E = E_k + E_p$

Kinetic energy $= \frac{1}{2} m\omega^2(r^2 - x^2)$

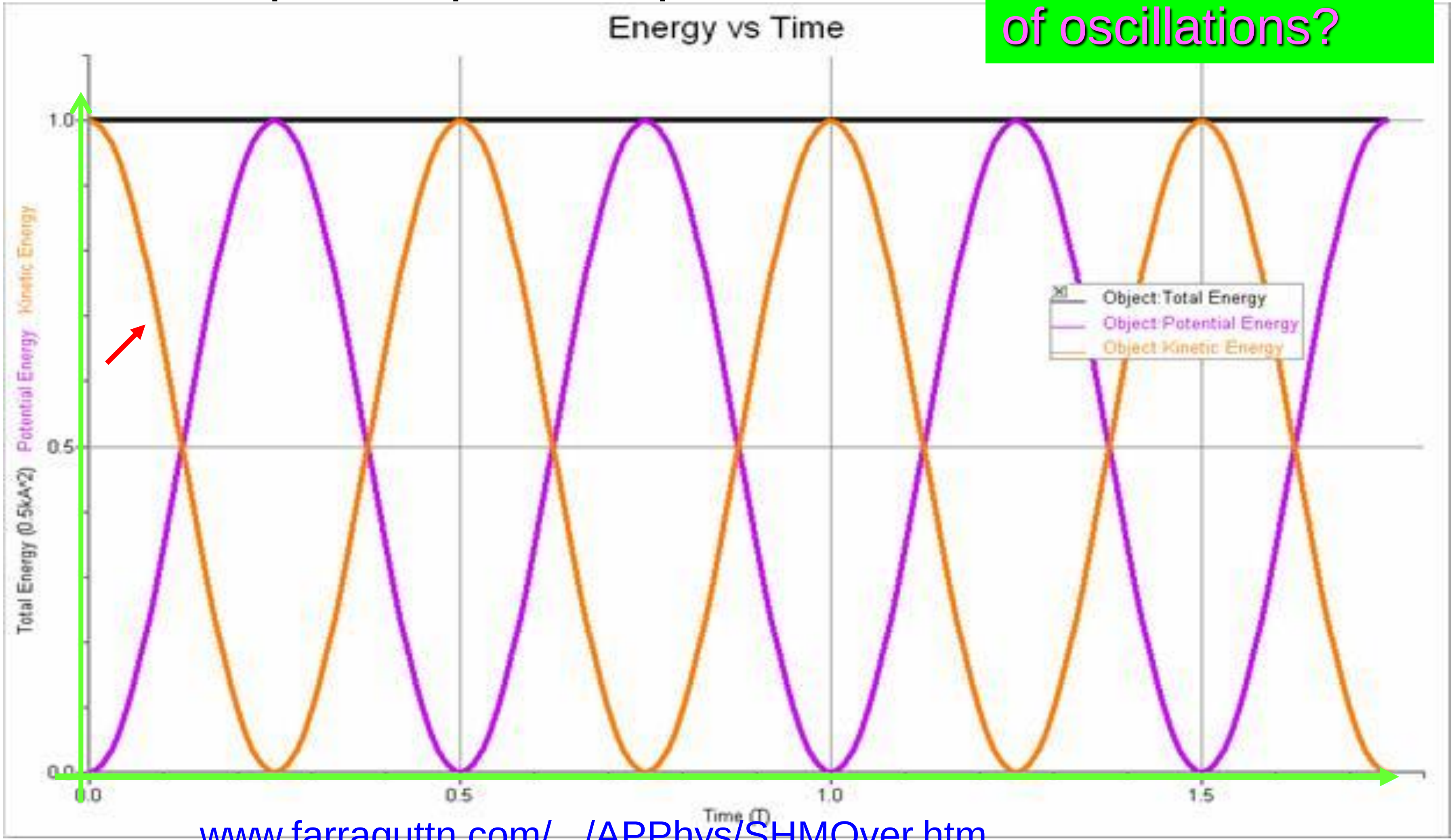
Potential energy $= \frac{1}{2} m\omega^2 x^2$

Total energy $= \frac{1}{2} m\omega^2 r^2$

Energy changes with time



What is the period of oscillations?



www.farraguttn.com/.../APPhys/SHMOver.htm

Energy in SHM

- E_K is due to the motion of mass.
- At maximum displacement E_K is zero
- At equilibrium E_K is maximum.

$$E_{K\,\max} = \frac{mv_{\max}^2}{2} = \frac{mr^2\omega^2}{2}$$

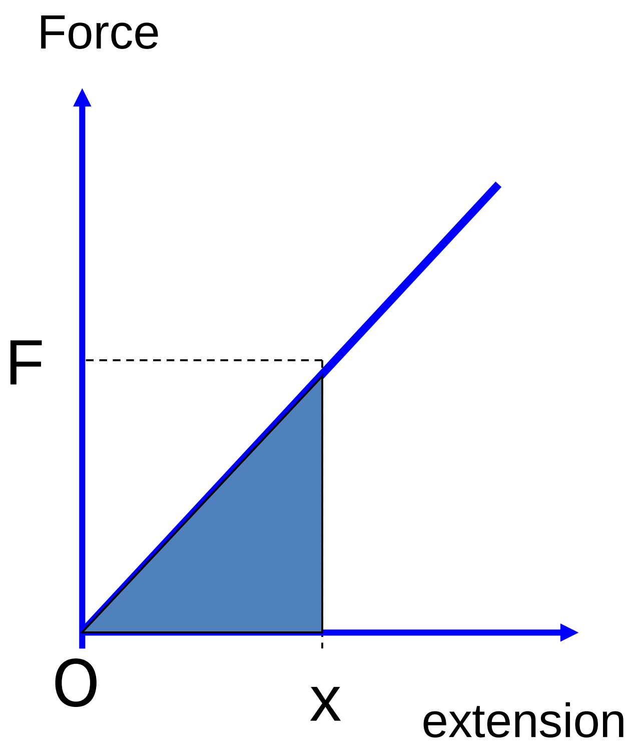
r = Amplitude

- E_P is due to position of mass from its equilibrium.
- At maximum displacement E_P is maximum.
- At equilibrium E_P is zero.
- E_P max $= \frac{1}{2}\, mr^2\, \omega^2$

$E_T = E_K + E_P = \frac{1}{2}\, mr^2\, \omega^2$

*To calculate E_P at any position; $E_P = E_T - E_K$

Review: Energy stored in spring



Potential energy stored in spring

= work done in stretching spring

= force x distance moved in direction of force

= shaded area

$W = \frac{1}{2} Fx$ or as $F = kx$

$= \frac{1}{2} kx^2$

- strain energy stored
- elastic potential energy

(a) Calculate the gain in potential energy when a mass of 150 g is raised through 1.0 mm. [2]

PYP 14.1

(b) A simple pendulum consists of a light inextensible string to which is attached a bob mass 150 g. The variation of V_p, the potential energy, with x, the horizontal displacement of the bob, is shown in Fig. 12.

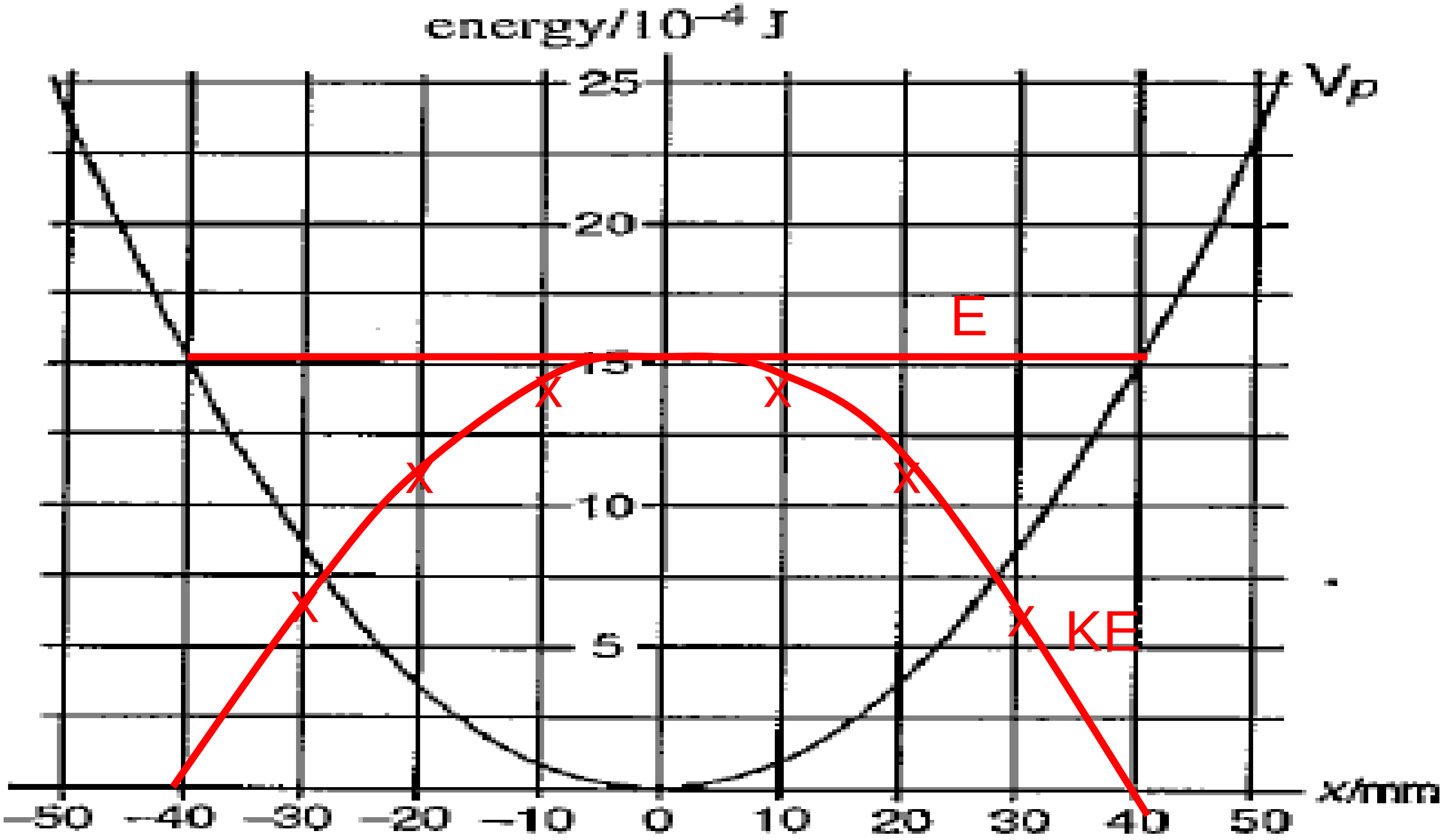


Fig. 12

In order to set the pendulum into oscillation, the bob is displaced sideways (keeping the string taut) until its centre of mass is raised vertically through 1.0 mm and then released. Using the axes of Fig. 12, sketch labelled graphs to show the variation, as the pendulum oscillates, of x with

(i) the total energy, (ii) the kinetic energy. [4]

(c) By reference to Fig. 12, or otherwise, write down the amplitude of oscillation of the pendulum. [2]

a) $mgh = 0.15(9.81)(0.001) = 14.7\text{x}10^{-4}$ J

c)40mm

oscillations

(a) free oscillations – simple harmonic motion with a constant amplitude and period and no external influences.

(b) damped oscillations – simple harmonic motion but with a decreasing amplitude due to external or internal damping forces.

(c) forced oscillations – simple harmonic motion but driven externally.

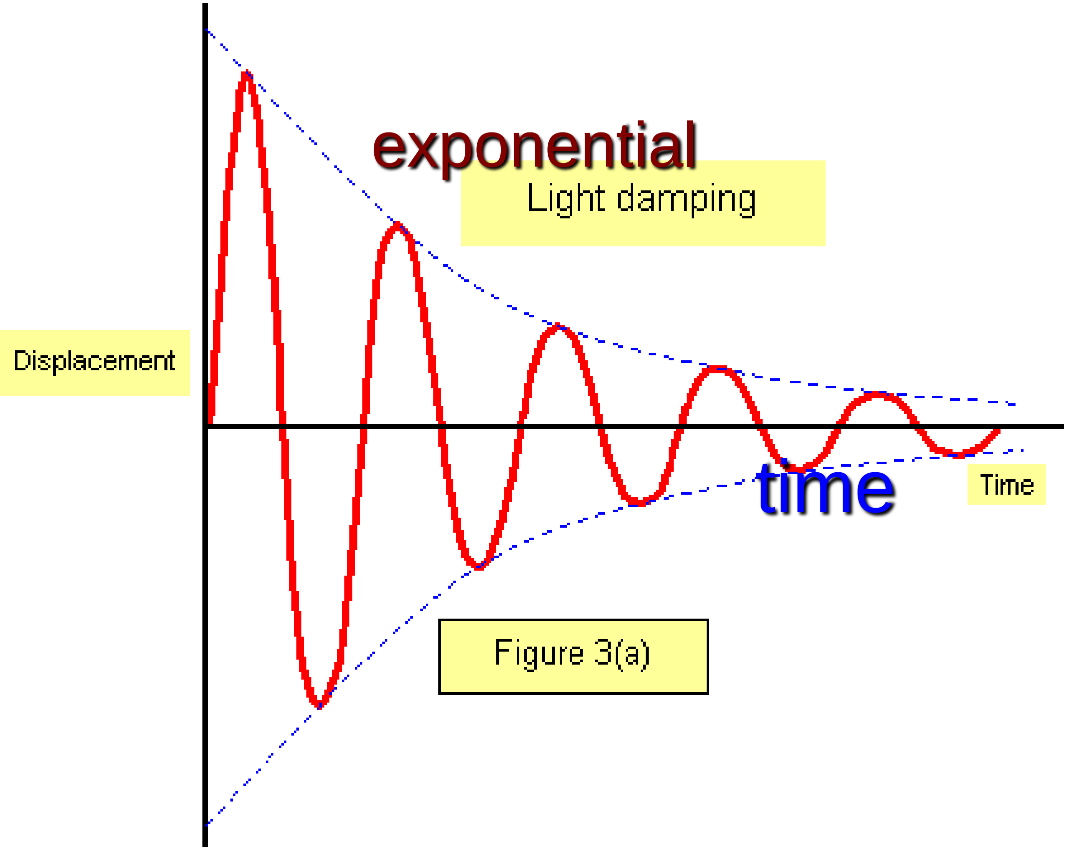
Free oscillations

- The amplitude remains constant as time passes, there is no damping.
- This type of oscillation will only occur in theory since in practice there will always be some damping.
- The displacement will follow the formula

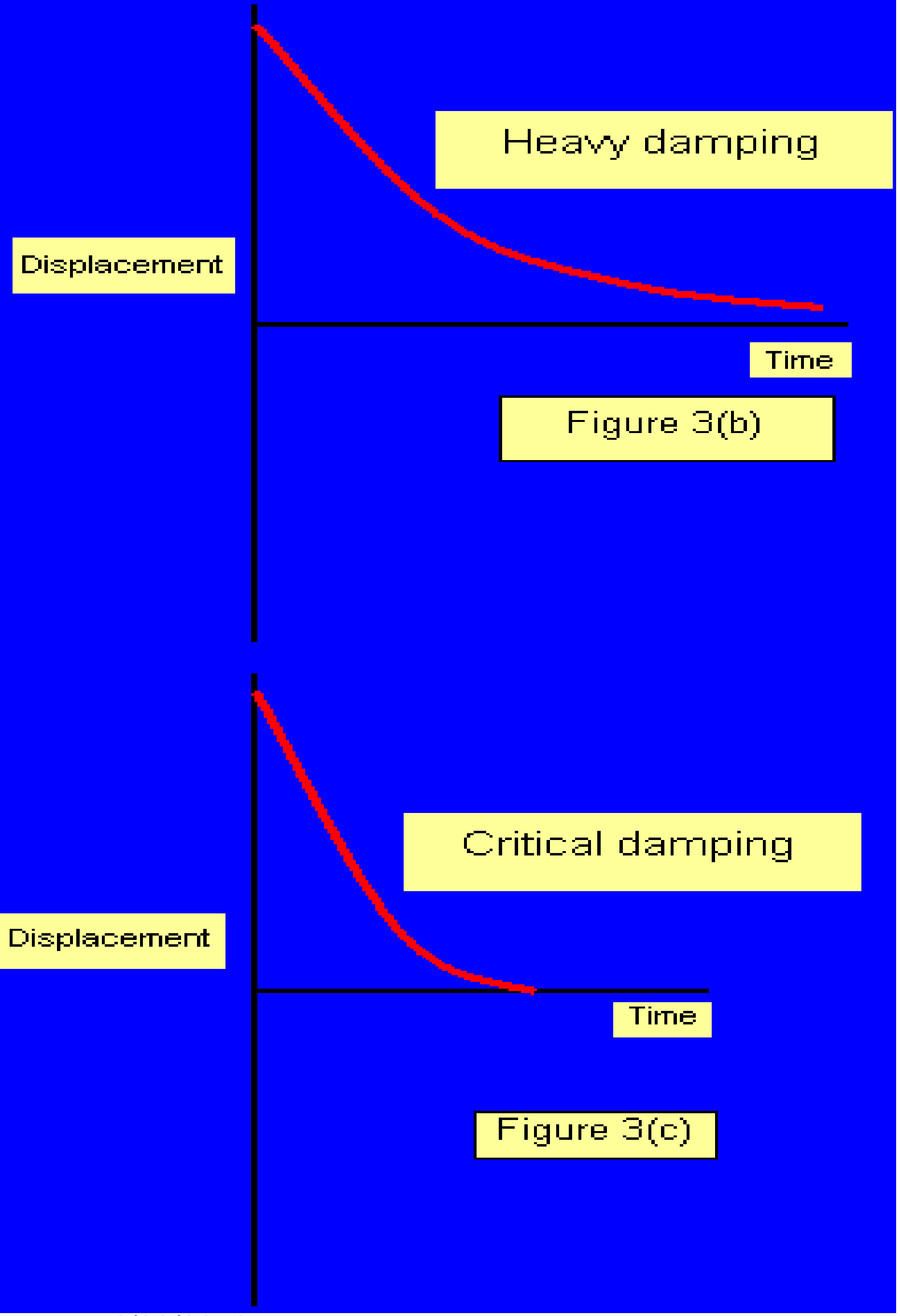
  $x = r \sin\omega t$ where r is the amplitude.

It is these types of oscillation that we have looked at already.

Damped oscillations



- Normally resistive forces are present to damped the motion.
- In air the resistive force is air resistance or friction.
- Mechanical energy is transformed to internal energy of the air molecules.
- Amplitude of oscillations decreases.



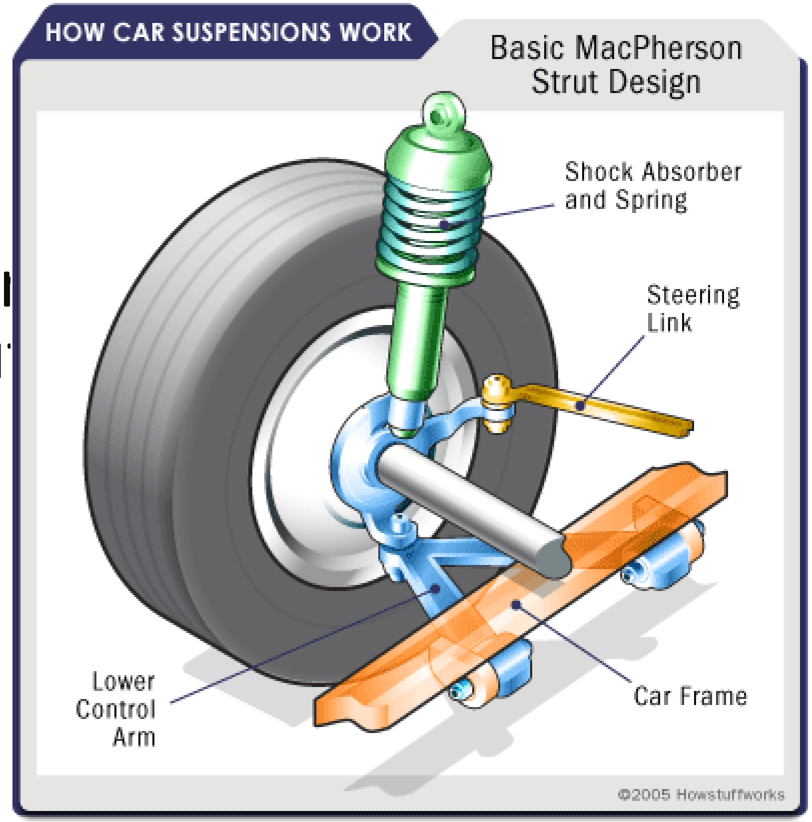
Heavy and critical damping

***Overdamped or heavy damping*.**

- Here the oscillating object is placed in very viscous medium.
- no oscillations occur.
- the object displaced take a long time to return to rest position.
- No useful applications.

Critically damped

- the object displaced, when r equilibrium position withou the shortest possible time.
- no oscillation occurs



Uses:

- moving coil meters,
- suspension system of car. A good car suspension is one in which the damping is lightly under critical damping as this results in a comfortable ride and quickly leaves the car ready to respond to further bumps in the road.

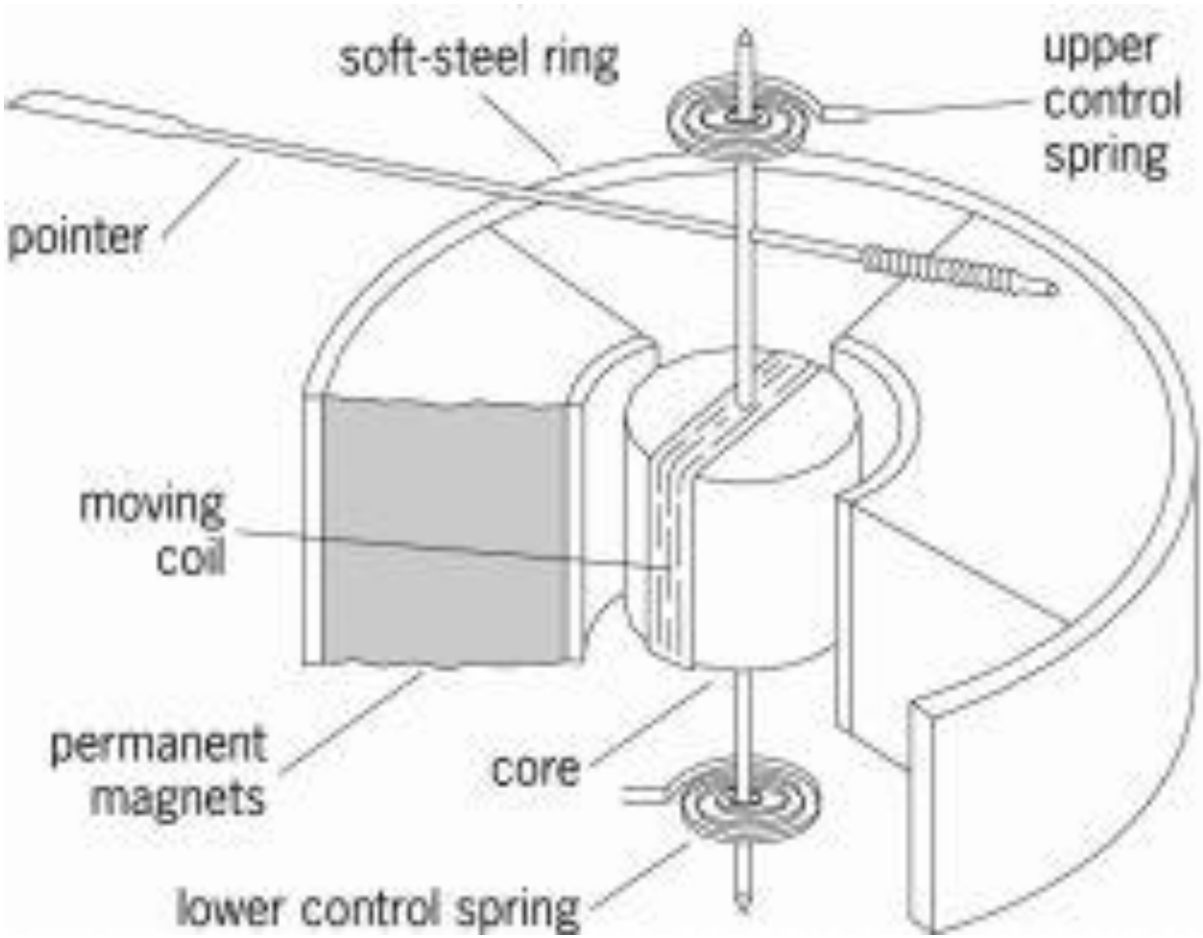
Damping

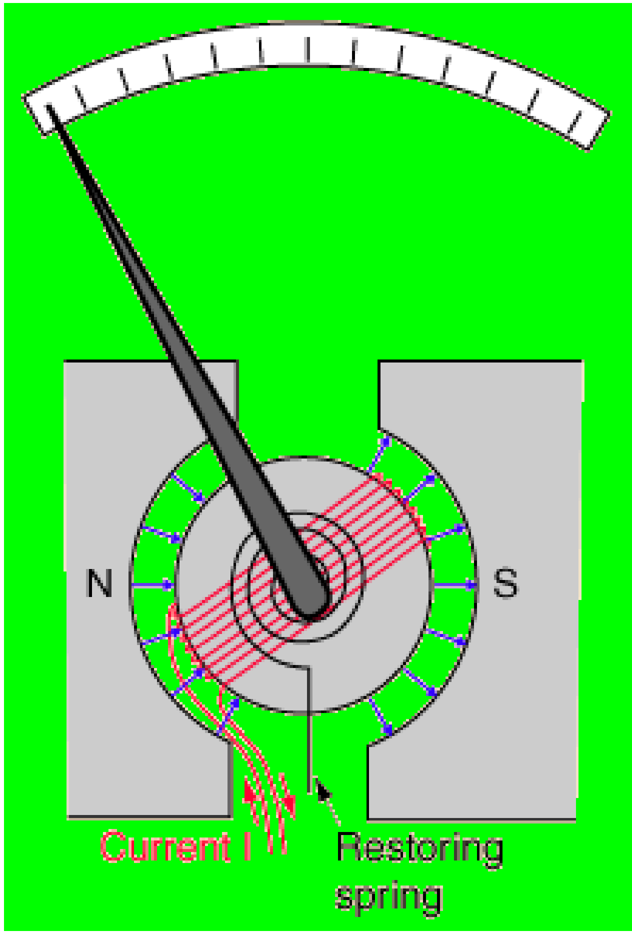
A good example of damping can be seen in the moving coil galvanometer. Electromagnetic damping is used here: the coil moves in a magnetic field and the current flowing in it can be shorted with a resistor, thus varying the damping.

The system is either
(i) dead beat — that is, critically damped, or
(ii) ballistic — the damping is as small as possible.

With reasonably light damping the period is unchanged but as the damping is increased the oscillations die away more rapidly.

moving coil galvanometer





Damping

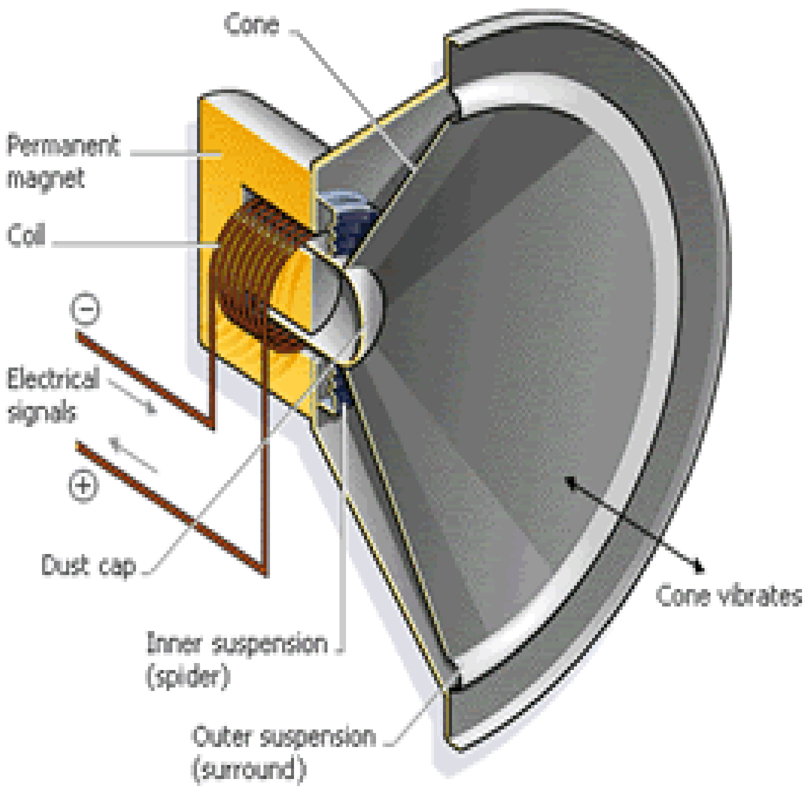
- Damping reduced the total mechanical energy of the oscillating system and thus the amplitude.
- Can we maintained the amplitude of the oscillator (keep amplitude constant)?

- Can we increased the amplitude of oscillations indefinitely?.

Forced oscillations

- These are vibrations that are driven by an period driving [external] force. A simple example of forced vibrations is a child's swing: as you push it the amplitude increases.
- A loudspeaker is also an example of forced oscillations; it is made to vibrate by the force on the magnet on the current in the coil fixed the speaker cone.

moving coil loudspeaker



- By varying the electric current through the wires around the electromagnet, the electromanget and the speaker cone can be made to back and forth. If the variation of the electric current is at the same frequencies of sound waves, the resulting vibration of the speaker cone will create sound waves, including that from voice and music.

http://www.school-for-champions.com/science/electromagnetic_devices.htm

Forced oscillations

- When the swing is displaced and released it oscillates with it natural frequency.
- Due to damping the amplitude of oscillations decreases.
- To keep the amplitude constant the driver must push at the right time that the swing is about to swing downwards.
- The driver provides the energy to the swing.

Forced oscillations

- When the frequency of the driver is the same as the natural frequency of the driven (oscillator) the oscillator oscillates with large amplitude.
- Damping or resistive forces limit the maximum amplitude.
- For low velocity the damping force is proportional to the speed, but for high speed the damping force is proportional to speed square.
- More energy needs to be provided by the driver to increase the amplitude of oscillations further.

Forced oscillation and Resonance

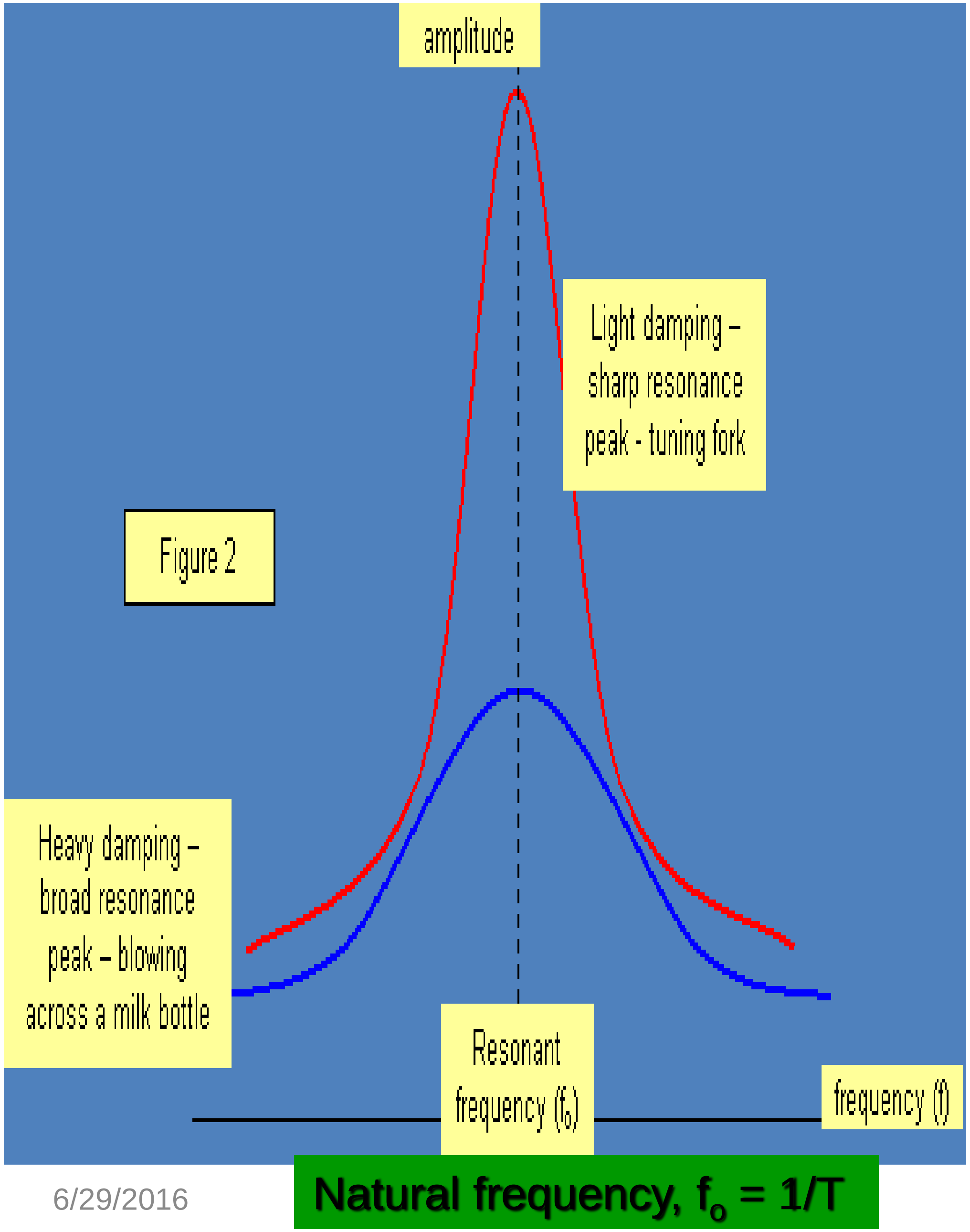
A ***forced oscillation*** occurs when a body is made to oscillate by the application of a periodic driving force i.e. a force applied at regular intervals.

At ***resonance*** the frequency (f) of the periodic driving force equals to the natural frequency of the body being forced to oscillate.

$f = f_o, 2f_o, \ldots$ or $\frac{1}{2}f_o$.

- The oscillator oscillates with maximum amplitude.
- The driving force supplied the energy for the oscillator to vibrate with maximum amplitude.
- The constant amplitude can also be maintain by half the natural frequency i.e. pushing the swing in alternate periodic interval.

Resonance



- Forced vibrations can also show another very important effect.
- With the swing you will find that if you push in time with the natural frequency of the swing then the oscillations build up rapidly.
- This last fact is an example of **resonance**.

Resonance

- The sharpness of the resonance depends on the amount of damping, being sharp for light damping.
- For heavy damping, the amplitude of oscillation at all frequencies is reduced and the peak becomes flatter.

Benefits

- ***Musical instruments*** rely on resonance to amplify the sound produced.
- Resonant vibration of quartz crystals are used to control clocks and watches.
- ***Electrical resonance*** occurs when a radio circuit is tuned by making its natural frequency for electrical oscillations equal to that of the incoming radio signal.
- **Telecommunication.** The electrons in a radio receiving aerial are forced to vibrate by the radio wave passing the aerial. If the aerial is the correct length for the particular frequency being used, then the amplitude of the oscillation is larger. So a large signal is passed by the aerial to the radio, where the circuitry again used

Troublesome

- Soldiers need to break step when crossing certain ***suspended wooden bridges***. Failure to do so cause the loss of over two hundred French infantryman in 1850.
- ***Opera singers*** can shatter wine glasses by forcing them to vibrate at their natural frequencies.
- ***Tacoma Narrows bridge*** disaster of 1940 was caused by the bridge being too slender for the wind conditions in the valley. One day strong winds set up twisting vibrations and the amplitude of vibration increased due to resonance, until eventually the bridge collapsed

Tacoma Narrows bridge

The original Tacoma Narrows Bridge opened on July 1, 1940. It received its nickname "Galloping Gertie" due to the vertical movement of the deck observed by construction workers during windy conditions. The bridge collapsed into Puget Sound the morning of November 7, 1940, under high wind conditions. Engineering issues as well as the United States' involvement in World War II postponed plans to replace the bridge for several years until the replacement bridge was opened on October 14, 1950.

Tacoma Narrows bridge

The original Tacoma Narrow Bridge, at all stages of its short life, was very active in the wind. Its nickname of *Galloping Gertie* was earned from its vertical motions in even very modest winds. Its collapse on November 7, 1940 attracted wide attention at the time and ever since, due in part to its capture on film.

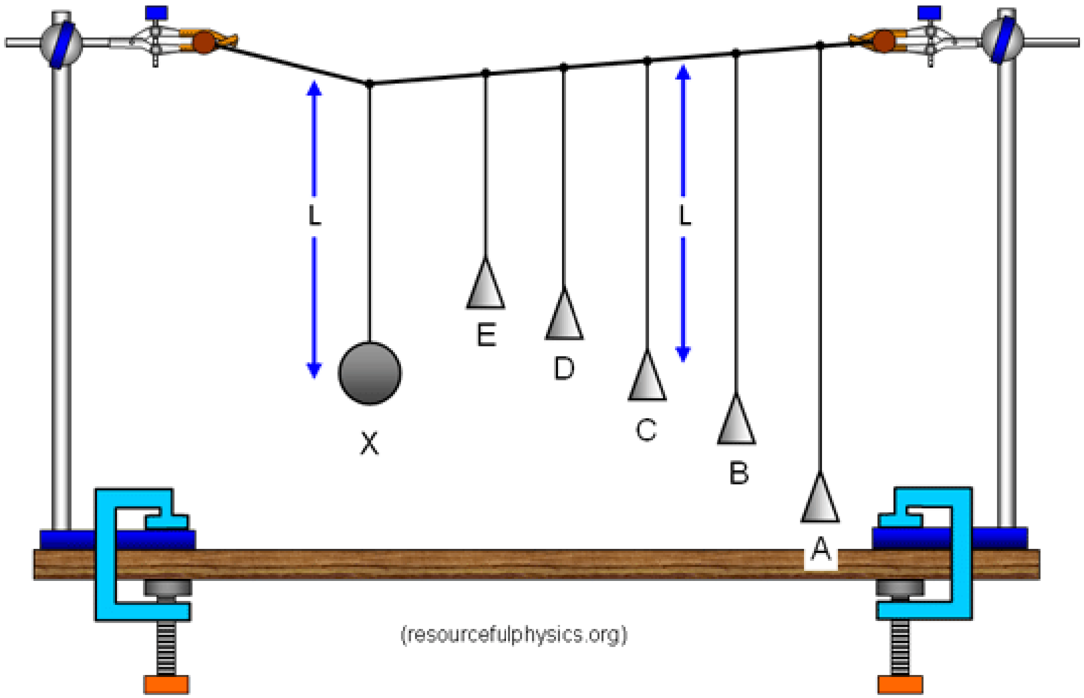
http://www.ketchum.org/bridgecollapse.html

Tacoma Narrows bridge

The bridge's collapse had a lasting effect on science and engineering. In many physics textbooks, the event is presented as an example of elementary forced resonance with the wind providing an external periodic frequency that matched the natural structural frequency, though its actual cause of failure was aeroelastic flutter.[1] Its failure also boosted research in the field of bridge aerodynamics-aeroelastics, the study of which has influenced the designs of long-span bridges built since 1940.

http://en.wikipedia.org/wiki/Tacoma_Narrows_Bridge_(1940)

Barton's pendulums.



- When pendulum X is displaced all the other pendulum move at different frequency.

  The pendulum having the same length as pendulum X will oscillate in phase with large amplitude.

  The cone allow the amount of damping to be increased.

www.ioppublishing.com/.../SHM/page_4481.html

Self Test 14

1) Define simple harmonic motion.
2) What is the phase difference between displacement and velocity in SHM?
3) The displacement x of a particle at time t is given by x/m = 5 sin (2t/s). what is
a) amplitude,
b) period

1)

2) ½ π radian

3a) 5 m

b) compare with equation

$x = x_o \sin \omega t$

$\omega = 2$ rad/s

$T = 2\pi/ \omega = 2\pi/ 2$

$= 3.14$ s

4) A body in simple harmonic motion makes n complete oscillation in one second. The angular frequency of this motion is ………..

5) What is the frequency of a SHM in which the acceleration is related to the displacement x by the equation $a = -\omega^2 x$?

6) The cone of a loudspeaker sounding a note of frequency f executes SHM of amplitude a. What is the maximum acceleration of the cone?

4) $\omega = 2\pi f = 2\pi n$

5) $f = \omega/2\pi$

6) $\mathbf{a} = -\omega^2.\mathbf{x}$

$= 4\pi^2 f^2 a$

Self Test 14.1

1) What is a damped motion?
2) What is a forced oscillation?
3) What is resonance?
4) Give practical application of resonance and a trouble some nature of resonance.
5) Give a practical application of critical damping.

1) One where the oscillating body is opposed by friction.
2) when an oscillating body is forced to oscillate by a periodic driving force
3) It occur when the frequency of the driver is the same as the oscillating system, the system oscillates with large amplitude.

5) car suspension system or shock absorber.

Simple Harmonic Motion

Amplitude: max. displacement from the rest position

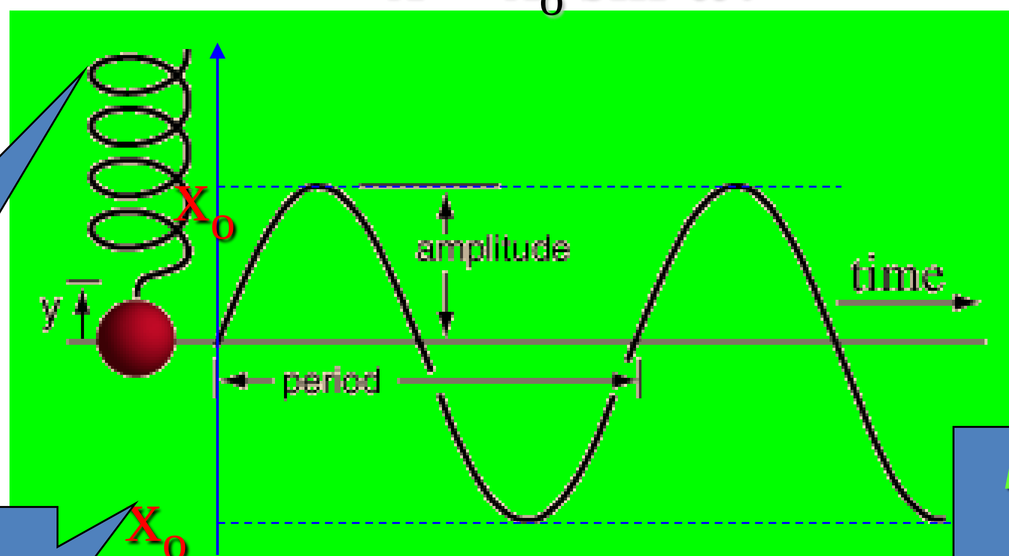
angular frequency is the frequency express in rad/s

$$x = x_o \sin \omega t$$

Period (T) of an oscillation is the time to complete one oscillation.

$$\omega = \frac{2\pi}{T}$$

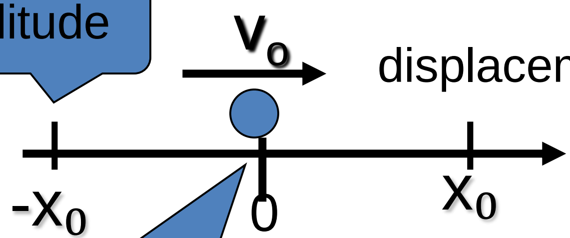
v and x are ½ π radian



a and x are π radian

$$\omega = 2\pi f$$

Frequency (f, ν (Gk nu)) of oscillation is the number of complete oscillations per unit time.



amplitude

Rest position or fixed point

Oscillations

Oscillations

oscillations and vibrations

Simple Harmonic Motion [SHM]

Oscillating simple pendulum
Mass attached to spring vibrating vertically

Definition
$\mathbf{a} = - \omega^2 . \mathbf{x}$

a= acceleration
x= displacement
acceleration is always opposite to displacement (negative)

ω= angular frequency (freq. express in rad/s)

$x = x_o \sin \omega t$

absent of resistive forces,
amplitude constant,
Total energy constant

forced oscillation

oscillator forces to oscillate with application of periodic driving force

Resonance occur when frequency of driver equal to natural f of oscillator

oscillator oscillates with large amplitude

Damped oscillation

presence of friction or resistive forces

total mechanical energy decreases, amplitude decreases